

A MERGED RESEARCH REPORT

Infinitesimal Analytic Geometry

over the Surcomplex Numbers

Radius-Free Germs, a Nullstellensatz,
and Finite Maps

September 21, 2026

Abstract

We develop the several-variable analytic geometry of the infinitesimal neighbourhood of an ordinary point in the surcomplex numbers $\mathbf{SC} = \mathbf{No}[i]$, and we do so over *two* coefficient rings whose theorems look alike and are not interchangeable. Fix a nonzero divisible set-sized ordered subgroup $\Gamma \subset \mathbf{No}$ and put $K = \mathbb{C}((t^\Gamma))$. The *radius-free* ring is $\mathcal{A}_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$, whose Hahn coefficients are individually convergent germs subject to no common radius; the *common-domain* ring $\mathcal{R}_n$ consists of germs of Hahn families all of whose coefficients are holomorphic on one ordinary neighbourhood. We prove $\mathcal{R}_n \subsetneq \mathcal{A}_n \subsetneq \mathbb{C}[[z]]((t^\Gamma))$, with explicit witnesses in both directions, and we prove separately of each of $\mathcal{A}_n$ and $\mathcal{R}_n$ that it is a Noetherian Jacobson domain of dimension n whose maximal ideals are exactly the infinitesimal evaluation ideals $\mathfrak{m}_a = (z_1 - a_1, \dots, z_n - a_n)$, $a \in \mathfrak{m}_K^n$, with residue field K , regular local rings, and completions $K[[X_1, \dots, X_n]]$. Neither ring is local: each records every infinitesimal point at once.

Over $\mathcal{A}_n$ we prove support-controlled Weierstrass division and preparation, the strong Nullstellensatz $I(Z(I)) = \sqrt{I}$ on the monad $\mathfrak{m}_K^n$ — with an example showing that the nonzero-value-group hypothesis is essential — invariance of a finite zero scheme under enlargement of the workspace, a full-surcomplex corollary accounting for every infinitesimal zero in the proper class $\mathbf{SC}$, faithful flatness of the inclusion into the formal-coefficient ring, and a sharp inverse-Jacobian Hensel–Rouché theorem with root matching. Over $\mathcal{R}_n$ we prove parameterized preparation and division on one fixed domain, Noether normalization, finite surjective projections of irreducible monad zero sets with generically reduced fibres of the expected degree, vanishing of all higher homology of the Hahn Koszul complex, and a base-change theorem. Both rings carry conservation of the finite length of an arbitrary isolated complete intersection under positive-support perturbation, a perfect residue pairing that survives collisions, and a trace–Jacobian identity; the two residue functionals, one defined by a strongly summable series and one by expansion on a fixed ordinary cycle, are shown to agree where both are defined. Worked examples include a four-point system splitting across different valuation ranks, a genuinely radius-free two-variable system with infinite individual residues summing to an infinitesimal, and a length-five collision with the exact collision equation $3125u^6 - 108s^5 = 0$.

Status. Every assertion below is proved from explicitly identified classical inputs. Multivariable Hahn–Weierstrass theory and non-Archimedean Nullstellensatz results have substantial published predecessors and are not claimed here as new general ideas. The proposed contribution is the precise combination: the radius-free coefficient ring with its structure package and workspace theorems, the common-domain normal-form and residue package with explicit support certificates, and the comparison between them. The literature search was made on 21 September 2026 and was targeted rather than exhaustive. No named published conjecture is claimed solved, no exhaustive priority certification is asserted, and none of the proofs has been formally verified. Symbolic checks validate the displayed examples only; they are not machine verification of the general proofs.

Contents

1	The problem, the two rings, and the contribution	1
1.1	From one moving zero to an analytic zero scheme	1
1.2	The four coefficient rings, and why they must be kept apart	1
1.3	Overview of the main statements	2
1.4	Predecessors and the status of novelty	3
1.5	Notation fixed once	4
2	Hahn workspaces and the summability mechanism	4
2.1	Set-sized workspaces inside the surcomplex class	4
2.2	Strong summability and the support calculus	5
2.3	The two negative companions	6
3	The coefficient rings and their values	7
3.1	The radius-free ring $\mathcal{A}_n$	7
3.2	The common-domain ring $\mathcal{R}_n$	8
3.3	The two inclusions are strict	9
3.4	Taylor calculus on the monad	10
4	Weierstrass division and preparation, over each ring	10
4.1	Division over the radius-free ring $\mathcal{A}_n$	10
4.2	Division over the common-domain ring $\mathcal{R}_n$	12
5	Noetherian geometry and the Nullstellensatz over $\mathcal{A}_n$	13
5.1	Finite generation	13
5.2	Every maximal ideal of $\mathcal{A}_n$ is an evaluation ideal	13
5.3	The strong theorem	14
5.4	Regularity, completion, and two warnings	14
6	Noetherian geometry, normalization and finite maps over $\mathcal{R}_n$	15
7	Finite maps of monad zero sets over $\mathcal{R}_n$	17
8	Isolated complete intersections: the classical ingredients	18
8.1	A linear division splitting	18
8.2	Local residues and duality, imported	19
8.3	The perturbation residue and its normalization	19
9	The deformation theorem over each ring	20
9.1	The radius-free form, over $\mathcal{A}_n$	20
9.2	Multiplication matrices and residue reconstruction over $\mathcal{A}_n$	21

9.3	The common-domain form, over $\mathcal{R}_n$: a Koszul conjugacy	22
9.4	A coefficient recursion, not an iteration limit	24
10	Residue duality, traces, and exact zero multiplicities	25
10.1	The contour-expansion functional over $\mathcal{R}_n$	25
10.2	Finite-parameter testing without a uniform radius	26
10.3	Actual points account for the whole length	27
11	Workspace invariance and the analytic–formal comparison	29
11.1	No additional roots at unrepresented surreal scales	29
11.2	Allowing divergent ordinary coefficient series	30
12	Quantitative Hensel–Rouché stability over $\mathcal{A}_n$	31
12.1	A local estimate at a simple root	31
12.2	Correspondence for an entire finite cluster	33
13	Worked examples	34
13.1	Four coupled roots and a perfect residue matrix, over $\mathcal{A}_2$	34
13.2	A genuinely radius-free, higher-rank deformation over $\mathcal{A}_2$	35
13.3	A length-five collision with three scale regimes, over $\mathcal{R}_2$	37
13.4	Beyond polynomial and grid-supported perturbations, over $\mathcal{R}_2$	39
14	Hypothesis audit, limits, and open questions	40
14.1	Hypotheses that cannot be dropped silently	40
14.2	What is proved, and what is not	41
14.3	Specific research directions left open	42
A	The finite-family classical input and the trace proof	43
B	Support accounting and proof dependencies	44
C	Notation	45
	References	46

1 The problem, the two rings, and the contribution

1.1 From one moving zero to an analytic zero scheme

Two supplied manuscripts on surcomplex analysis construct one-variable preparation and moving-zero residue theory on infinitesimal thickenings of ordinary complex domains, and both identify several-variable division and finite mappings as their principal further targets [6, 7]. The dependence of the present report on them is motivational, not logical: none of their unproved targets is used here as a theorem.

The several-variable question is not whether a simple root lifts. The implicit function theorem already answers that. The substantial question is whether a *singular* system in several variables retains its entire local algebra when its equations acquire arbitrary infinitesimal Hahn perturbations. The ordinary system (x^2, y^2) has a zero of length four; an infinitesimal coupling can split it into four points lying at different coordinate scales. Counting only the ordinary leading zero sees one point; counting only distinct perturbed zeros misses collisions; and an argument that treats all infinitesimals as powers of a single real-valued parameter can miss entire valuation ranks. The appropriate invariant is the dimension of the finite quotient algebra, together with its residue pairing and its multiplication operators.

A second issue arises at the same moment, and it is the issue that makes this report a report about two rings rather than one. Infinitely many ordinary analytic coefficients may have radii of convergence tending to zero. Requiring one common ordinary polydisk excludes them, even though each coefficient has an unambiguous Taylor expansion at an infinitesimal argument. Dropping the requirement admits them, but then no fixed ordinary domain is available on which to run a Cartan-theoretic or contour argument. Both choices are legitimate, both support a structure theory, and the theorems they support are *not* the same theorems, because they are theorems about different rings.

1.2 The four coefficient rings, and why they must be kept apart

Throughout, $\Gamma \subset \mathbf{No}$ is a nonzero divisible set-sized ordered subgroup, $K = \mathbb{C}((t^\Gamma))$, $\mathfrak{m}_K$ is its ideal of infinitesimals, and $R_n = \mathbb{C}\{z_1, \dots, z_n\}$ is the ring of convergent germs at the origin. Four coefficient rings occur in this subject under closely similar notation, and all four carry closely similar structural statements:

1. the *fixed-polydisk* ring $\mathcal{O}(D)((t^\Gamma))$, for one ordinary polydisk D held fixed once and for all;
2. the *common-domain germ ring* $\mathcal{R}_n = \varinjlim_{U \ni 0} \mathcal{H}_\Gamma(U)$, where $\mathcal{H}_\Gamma(U)$ consists of Hahn families all of whose coefficient functions are holomorphic on the one ordinary neighbourhood U , and the limit is over the restriction maps;
3. the *radius-free* ring $\mathcal{A}_n = R_n((t^\Gamma))$, whose Hahn coefficients are individually convergent germs with no common domain required;
4. the *formal-coefficient* ring $\mathcal{F}_n = \mathbb{C}[[z_1, \dots, z_n]]((t^\Gamma))$.

There are natural maps

$$\mathcal{O}(D)((t^\Gamma)) \longrightarrow \mathcal{R}_n \subsetneq \mathcal{A}_n \subsetneq \mathcal{F}_n, \quad (1.1)$$

and the two inclusions displayed as strict are proved strict in [theorems 3.10](#) and [3.11](#). Each of the four rings satisfies, *of itself*, a statement of the form “the ring is Noetherian; its maximal ideals are the infinitesimal evaluation ideals; its local rings are regular with completion $K[[X]]$; division holds with support in $S + E^*$.” The statements are nearly word for word identical and the notation nearly

collides. This is a definitional divergence that reads like a contradiction and is not one; it is also the single likeliest place in this subject to produce a false attribution. We therefore adopt, and keep, the following discipline.

Convention 1.1 (Ring discipline). Every numbered statement below names the ring over which it is asserted. No theorem proved of one of the four rings is transferred to another. Where a statement holds of two rings, it is stated twice, with the two proofs given and compared, because the proofs differ in an essential resource: over $\mathcal{A}_n$ one may use ordinary germ-level division maps but has no fixed ordinary domain; over $\mathcal{R}_n$ one has a fixed ordinary domain and may use Cartan's theorem B and an ordinary residue cycle on it, but one cannot reach a coefficient family with shrinking radii. We do not unify the notation into a single symbol.

The two results that relate the rings are the ones to invoke instead of a transfer. The first is [theorem 3.10](#): the element

$$F(z) = \sum_{r \geq 1} \frac{t^{r\eta}}{1 - rz} \in \mathcal{A}_1$$

has a pole at $1/r$ in its coefficient at $r\eta$, so it lies in no $\mathcal{H}_\Gamma(U)$ and is not an element of $\mathcal{R}_1$ with those coefficients. Consequently invoking a theorem stated only for coefficients holomorphic on one fixed polydisk would leave a gap in the radius-free setting. The second is [theorem 11.5](#): the inclusion $\mathcal{A}_n \hookrightarrow \mathcal{F}_n$ is faithfully flat, with the same maximal ideals, residue fields, completions and finite-colength quotients — so the passage from convergent to formal coefficients, unlike the passage from common-domain to radius-free coefficients, changes nothing about a finite zero scheme.

1.3 Overview of the main statements

The following four theorems summarize the report. Each names its ring.

Theorem 1.2 (Structure of the radius-free ring $\mathcal{A}_n$, overview). *The ring $\mathcal{A}_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$ is a regular Noetherian Jacobson domain of dimension n . There is a bijection*

$$\mathfrak{m}_K^n \longrightarrow \text{Max}(\mathcal{A}_n), \quad a \longmapsto \mathfrak{m}_a = (z_1 - a_1, \dots, z_n - a_n),$$

every residue field is K , for every ideal $I \subseteq \mathcal{A}_n$ one has $I(Z_{\mathfrak{m}_K^n}(I)) = \sqrt{I}$, and the completion of $(\mathcal{A}_n)_{\mathfrak{m}_a}$ is $K[[z - a]]$. The ring $\mathcal{A}_n$ is not local.

Theorem 1.3 (Structure and finite maps for the common-domain ring $\mathcal{R}_n$, overview). *The ring $\mathcal{R}_n$ is a Noetherian Jacobson domain of dimension n with the same description of its maximal ideals, residue fields, local rings and completions. In addition, every quotient $\mathcal{R}_n/\mathfrak{p}$ by a prime of dimension d is finite over a coordinate copy of $\mathcal{R}_d$; the induced projection $V(\mathfrak{p}) \rightarrow \mathfrak{m}_K^d$ of monad zero sets is surjective with finite fibres, and over the complement of one nonzero $s \in \mathcal{R}_d$ every fibre consists of exactly $r = [\text{Frac}(\mathcal{R}_n/\mathfrak{p}) : \text{Frac}(\mathcal{R}_d)]$ distinct reduced points.*

Theorem 1.4 (Conservation, duality and the radius-free strengthening, overview). *Let $f_1, \dots, f_n \in R_n$ vanish at the origin with $\mu = \dim_{\mathbb{C}} R_n/(f_1, \dots, f_n) < \infty$, and let $b_1, \dots, b_\mu$ be monomials whose classes form a basis of $B_0 = R_n/(f)$. Put $F_i = f_i + E_i$ with $v_H(E_i) > 0$.*

1. *If the E_i lie in $\mathcal{R}_n$, then the classes of the b_r are a K -basis of $A_F = \mathcal{R}_n/(F)$, all higher homology of the Hahn Koszul complex of F over $\mathcal{R}_n$ vanishes, and all normal-form and division outputs on one fixed ordinary domain have support in $\text{supp } G + E^*$.*

2. If the E_i lie in the strictly larger ring $\mathcal{A}_n$, the same basis statement and the same support certificate hold for $B_F = \mathcal{A}_n/(F)$, by a different proof. In both cases $\dim_K B_F = \mu$, the associated residue pairing is perfect, $\text{Res}_F(HJ_F) = \text{Tr}_K(M_H)$ and $\text{Res}_F(J_F) = \mu$, the zero set $Z_F \subset \mathfrak{m}_K^n$ is finite and nonempty with $\sum_{a \in Z_F} e_a = \mu$, and enlarging the workspace Γ changes neither the zeros nor their multiplicities.

Theorem 1.5 (Quantitative stability over $\mathcal{A}_n$, overview). *Let $F \in \mathcal{A}_n^n$ have $v_H(F_i) \geq 0$ and let $a \in \mathfrak{m}_K^n$ be a simple zero. Put $\kappa_a = \max(0, -\min_{i,j} v((DF(a)^{-1})_{ij}))$. If $G = F + D$ and $\delta = \min_i v_H(D_i) > 2\kappa_a$, there is exactly one zero b of G with $v_{\min}(b - a) > \kappa_a$; it is simple and satisfies $v_{\min}(b - a) \geq \delta - \kappa_a$. The strict factor two cannot be weakened.*

Theorem 1.2 is proved in sections 4 and 5; theorem 1.3 in sections 6 and 7; theorem 1.4 in sections 9 to 11; and theorem 1.5 in section 12. The maximal-ideal assertions are stronger than the mere ability to substitute infinitesimals: they say that substitution accounts for *all* closed points of these algebras.

1.4 Predecessors and the status of novelty

A literature check changes the appropriate framing of these results, and we adopt the more careful of the two framings available to us.

Cluckers–Lipshitz–Robinson prove multivariable preparation and division for nonstandard analytic structures [5, Theorem 2.5 and Remark 2.6]. Cluckers–Lipshitz explicitly treat analytic coefficient families with a *common radius* and arbitrary well-ordered Hahn supports [4, §3.1, Remark 3.1.4, Example 3.3(6)], and their separated analytic theory contains normalization and Nullstellensatz theorems [4, §5.2]. These are close and substantial predecessors. Passing to complex coefficients, phrasing their consequences in the terminology of the supplied manuscripts, or giving another Weierstrass recursion is not by itself a novelty claim. Neither a Hahn Weierstrass theorem nor a non-Archimedean Nullstellensatz is claimed here as a new general idea. The coefficient ring and its support restrictions must be compared, not just the theorem titles: our radius-free ring $\mathcal{A}_n$ retains convergence of each complex coefficient germ but imposes no common radius, and theorem 3.10 proves that it strictly contains the common-domain ring $\mathcal{R}_n$. We do not claim that $\mathcal{A}_n$ lies outside every broader analytic-structure framework.

Conservation of intersection number and residue duality are classical in complex analytic geometry [14, 13, 18, 12], and the perturbation mechanism used over $\mathcal{R}_n$ is a special case of homological perturbation theory [9]. Residue expansions, normal forms and trace identities have classical antecedents, notably Cattani–Dickenstein–Sturmfels [3], Scheja–Storch [18] and Kunz–Waldi [15]. Surreal analysis likewise has substantial antecedents: Alling develops analytic constructions over surreal number fields [1], Berarducci–Mantova study transseries as surreal function germs [2], and Costin–Ehrlich develop surreal extension and integration theory [8]. Our differentiation is differentiation in the variables z_i with Hahn coefficients held fixed; it is not a derivation on the field of surreal numbers.

The new-work claim is deliberately narrow. We give an explicit fixed-neighbourhood realization of the perturbation mechanism for arbitrary Hahn supports, retain a specified basis throughout the deformation, obtain uniform support certificates for normal forms and matrices, prove compatibility with all divisible workspace enlargements, identify the associated moving-root residue functional including its collision-stable pairing, and — separately, over the strictly larger radius-free ring — prove the structure package, the strong Nullstellensatz, the workspace and full-surcomplex theorems, the analytic–formal comparison, and the quantitative root correspondence. In the sources inspected

we did not locate these exact combined statements for either ring. We offer them as proposed new results in a precisely specified setting; they may also be derivable from a broader existing framework. This is a proposed original synthesis and formulation, not a certified first theorem in multivariable non-Archimedean analysis. The search was conducted on 21 September 2026 and was not a systematic review of all publications. The proofs, rather than a priority assertion, are the substantive deliverable.

1.5 Notation fixed once

The sources of this report disagree on several symbols and on one name. We fix the following and use nothing else.

- Convention 1.6* (Notation). 1. *Reduction*. The map $K^\circ \rightarrow \mathbb{C}$ extracting the coefficient of t^0 is written red, and the associated topology on $\mathfrak{m}_K^n$ obtained by pulling back the ordinary topology of $\mathbb{C}^n$ along standard parts is called the *reduction topology*. It is called the residue topology and the shadow topology elsewhere; those are synonyms for the same object and are not used again.
2. *Evaluation*. The value of F at an infinitesimal point a is written $F(a)$. The notation $F^\sharp(a)$ occurs in the sources for the same operation and is not used again.
3. *Valuations*. v is the field valuation on K , with $v(0) = +\infty$; v_H is the Hahn valuation $\min \text{supp}$ on a ring of Hahn series with coefficients in a coefficient ring; $v_{\min}(a) = \min_i v(a_i)$ for a tuple.
4. *Support monoid*. For well-ordered $E \subset \Gamma_{>0}$ we write $E^* = \{0\} \cup \{e_1 + \cdots + e_r : r \geq 1, e_j \in E\}$. The symbol T^* is used for the same object in one source and is not used again.
5. *Deformation data*. μ is the ordinary colength; $b_1, \dots, b_\mu$ the fixed monomial basis; NF_F the normal-form map with coordinates $\text{NF}_F(G)_r$; Q_j the division certificates; M_H the multiplication matrix of H ; $\mathcal{G}_F$ the residue Gram matrix; e_a the local algebraic multiplicity at a (written $\mu_a(F)$ in one source); J_F the Jacobian determinant; κ_a the inverse-Jacobian valuation loss.
6. *Residues*. res_g is the ordinary local Grothendieck residue. Res_F is the *series* functional (8.8), defined over $\mathcal{A}_n$; Λ_F is the *contour-expansion* functional (10.2), defined over $\mathcal{R}_n$. They are distinct definitions and are shown to agree where both apply (theorem 10.2).
7. $\mathbb{N}$ contains zero. $\mathbf{SC} = \mathbf{No}[i]$, and $t^\gamma = \omega^{-\gamma}$.

2 Hahn workspaces and the summability mechanism

This section states the foundations that recur in every later proof, in the one place where they belong. Everything in it is common to both coefficient rings; it is the last material in this report of which that is true.

2.1 Set-sized workspaces inside the surcomplex class

A surcomplex number is an element of $\mathbf{SC} = \mathbf{No}[i]$. We use the normal-form monomials $t^\gamma = \omega^{-\gamma}$, so that a positive exponent means an infinitesimal. The Conway normal-form theorem identifies $\mathbb{C}((t^\Gamma))$ with a subfield of $\mathbf{SC}$: its elements have well-ordered exponent support and complex coefficients. The Hahn-field theorem says that $K = K_\Gamma = \mathbb{C}((t^\Gamma))$ is algebraically closed when Γ is divisible and the coefficient field is algebraically closed of characteristic zero; see [17, Corollary 4]. Write

$$R_\Gamma = \mathbb{R}((t^\Gamma)), \quad K = R_\Gamma[i], \quad K^\circ = \{a : v(a) \geq 0\}, \quad \mathfrak{m}_K = \{a \in K : v(a) > 0\} \cup \{0\},$$

with R_Γ real closed, $\text{red} : K^\circ \rightarrow \mathbb{C}$ the reduction of [theorem 1.6](#), and $v_{\min}(a) = \min_i v(a_i)$ for tuples. Under the surreal embedding, $\mathfrak{m}_K$ consists exactly of the elements of K that are infinitesimal relative to the positive real numbers.

Lemma 2.1 (Enlargement rule). *Every finite tuple of surcomplex numbers lies in some set-sized algebraically closed workspace K_Δ of the above form: take the additive subgroup generated by the union of the normal-form supports of the entries, enlarge by a positive element if necessary, and pass to the divisible hull.*

This observation is what later turns a workspace theorem into a statement about all surcomplex zeros. Its limit must be stated with equal force: it never authorizes a proper-class sum or a proper-class polynomial ring. No proper-class construction is used anywhere below. Every theorem about a ring concerns a fixed set-sized Γ , and the passage to the full class is made only by the explicit base-change arguments of [section 11](#), for one finite cluster at a time.

2.2 Strong summability and the support calculus

Definition 2.2 (Strong summability and the monoid E^*). A family of Hahn series is *strongly summable* if the union of its supports is well ordered and only finitely many members contribute to each exponent; its sum is then defined coefficientwise. For a well-ordered $E \subset \Gamma_{>0}$,

$$E^* = \{0\} \cup \{e_1 + \cdots + e_r : r \geq 1, e_j \in E\}.$$

This is the only notion of infinite sum used in this report. It replaces any topological convergence, and no continuity or norm hypothesis is ever invoked in its place.

Lemma 2.3 (Hahn–Neumann support lemma). *Let S_1, S_2 be well-ordered subsets of an ordered abelian group. Then $S_1 + S_2$ is well ordered and every exponent has only finitely many representations $s_1 + s_2$ with $s_i \in S_i$. If $E \subset \Gamma_{>0}$ is well ordered then E^* is well ordered, and a fixed exponent has only finitely many representations as a finite word in E , including finitely many possible word lengths. Consequently, for well-ordered S , the data indexed by a member of S together with a word in E are locally finite over $S + E^*$.*

The empty word represents zero. Positivity of E is essential: zero increments would permit infinitely many contributions at one exponent. These facts hold verbatim for supports of vector-valued series. The ring construction and the geometric inversion are reviewed in [\[17, §3\]](#); the ordered-series support theorem originates in [\[16\]](#).

Definition 2.4 (Hahn families of vectors). For a complex vector space V , let $V((t^\Gamma))$ denote the well-ordered Hahn families with coefficients in V . A complex-linear map $L : V \rightarrow W$ acts coefficientwise and never increases support.

The last sentence carries more weight than its length suggests. It makes no continuity assumption on L , so ordinary holomorphic division maps may be applied to infinitely many coefficient functions even when those functions have arbitrarily large ordinary norms, and — over $\mathcal{A}_n$ — even when they share no analytic neighbourhood at all.

Lemma 2.5 (Positive-support operator inversion). *Let V be a complex vector space, let $E \subset \Gamma_{>0}$ be well ordered, and let*

$$T = \sum_{e \in E} t^e T_e$$

act on $V((t^\Gamma))$, each $T_e : V \rightarrow V$ complex-linear. Then

$$(\text{id} + T)^{-1}G = \sum_{r \geq 0} (-T)^r G \quad (2.1)$$

is strongly summable for every G , and defines a K -linear operator. If $S = \text{supp } G$, the output support lies in $S + E^*$. The same statement holds for finite direct sums of vector spaces and for matrices of such operators.

Proof. A contribution to the coefficient at ν in $T^r G$ is indexed by $s + e_1 + \cdots + e_r = \nu$ with $s \in S$ and $e_j \in E$. By [theorem 2.3](#) there are only finitely many such words, including finitely many possible lengths, so each output coefficient is a finite sum of compositions of the T_e applied to finitely many input coefficients. The union of the possible exponents lies in the well-ordered set $S + E^*$. Multiplication by $\text{id} + T$ telescopes coefficientwise to G on either side. Commuting an additional scalar Hahn series through these operators is justified by the same finite-contribution property, which gives K -linearity. A finite matrix index set changes nothing. $\square$

2.3 The two negative companions

Both facts in this subsection are load-bearing. Omitting either invites a proof that looks like an iteration argument and is not one.

Remark 2.6 (Strong summability is not a valuation limit). Take $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$ with its inherited order; for every natural number N one has $N < \omega$. The partial sums of $\sum_{r \geq 0} t^r$ therefore need not converge in the valuation topology: their tails never have valuation greater than ω . Nevertheless the family is strongly summable and its sum is $(1 - t)^{-1}$. The same phenomenon appears in $\mathbb{Q} \oplus \mathbb{Q}$ ordered lexicographically: with $e = (0, 1)$ and $b = (1, 0)$, every ne is below b , yet $\sum_{n \geq 0} t^{ne} = (1 - t^e)^{-1}$ while the errors never have valuation above b . Hence no argument of the form “the valuation improves by a fixed positive amount at each iteration” justifies any construction in this report. [Theorem 2.5](#) does justify them, and replacing it by an assertion that errors tend to zero would discard legitimate cases of every theorem below.

Remark 2.7 (Finitely many words at one exponent is not finitely many exponents below it). A well-ordered positive support may have infinitely many exponents below a prescribed bound. Our proofs never require those initial segments to be finite; they require finitely many *words* contributing to one exact exponent. For $E = \{2 - 1/m : m \geq 1\}$ there are infinitely many members below 2, yet the coefficient of a given Hahn product is still a finite convolution. The distinction is precisely what makes the finite-parameter-replacement arguments of [theorem 10.4](#) and [theorem 10.5](#) legitimate: at a prescribed exponent, only finitely many coefficient germs are involved, and that is all a comparison with an ordinary finite-parameter deformation needs. In $\mathbb{Q} + \mathbb{Q}\omega$ the assertion “only finitely many exponents below” is false below ω .

Remark 2.8 (Well-ordering cannot be weakened to positivity). The set $\{1/r : r \geq 1\}$ is positive but not well ordered, so the corresponding putative Hahn series is not an element of any of the four rings of [section 1.2](#). Positivity alone would break [theorem 2.3](#) and with it every support certificate below.

3 The coefficient rings and their values

3.1 The radius-free ring $\mathcal{A}_n$

Definition 3.1 (The radius-free Hahn-germ ring). Let $R_n = \mathbb{C}\{z_1, \dots, z_n\}$ and define

$$\mathcal{A}_n = R_n((t^\Gamma)) = \left\{ \sum_{\gamma \in S} f_\gamma(z) t^\gamma : S \subset \Gamma \text{ well ordered, } f_\gamma \in R_n \right\}. \quad (3.1)$$

For $F \neq 0$ put $v_H(F) = \min \text{supp } F$ and let $\text{lc}(F)$ be its leading coefficient germ. A hypothesis $v_H(F) > 0$ permits $F = 0$.

The symbol $\mathcal{A}_n$ does not mean an unspecified ring of convergent series over K : [theorem 3.1](#) is the entire convention. In particular no convergence condition on the numerical sizes of the complex Taylor coefficients is imposed uniformly in γ , and the radii of the f_γ may tend to zero. The Hahn construction makes $\mathcal{A}_n$ a K -algebra and an integral domain, since the leading coefficient of a product is the product of the leading coefficient germs in the domain R_n ; and $\mathcal{A}_0 = K$.

Proposition 3.2 (Units of $\mathcal{A}_n$). *For $F = t^\lambda(f_0 + E) \in \mathcal{A}_n$ with $v_H(E) > 0$ and $f_0 \neq 0$, the following are equivalent: F is a unit of $\mathcal{A}_n$; f_0 is a unit of R_n ; $f_0(0) \neq 0$.*

Proof. If f_0 is a unit then $F^{-1} = t^{-\lambda} f_0^{-1} \sum_{r \geq 0} (-f_0^{-1} E)^r$ is strongly summable by [theorem 2.5](#). Conversely if $FG = 1$ then the product of the leading coefficient germs is 1, so f_0 is a unit. The last equivalence is ordinary local analytic algebra. $\square$

Proposition 3.3 (Evaluation and translation over $\mathcal{A}_n$). *For $a \in \mathfrak{m}_K^n$ and $F = \sum f_\gamma t^\gamma \in \mathcal{A}_n$, the expression*

$$F(a) = \sum_{\gamma, \alpha} \frac{\partial^\alpha f_\gamma(0)}{\alpha!} t^\gamma a^\alpha \quad (3.2)$$

is strongly summable and defines a K -algebra homomorphism $\text{ev}_a : \mathcal{A}_n \rightarrow K$. The formula

$$\tau_a F(z) = \sum_{\gamma, \alpha} \frac{\partial^\alpha f_\gamma(z)}{\alpha!} t^\gamma a^\alpha \quad (3.3)$$

defines an automorphism of $\mathcal{A}_n$ with $\tau_a \tau_b = \tau_{a+b}$, $\tau_a^{-1} = \tau_{-a}$ and $(\tau_a F)(b) = F(a + b)$.

Proof. Let $S = \text{supp } F$ and let E_a be the union of the supports of the coordinates of a , a well-ordered positive set. Every expanded term in either formula has support in $S + E_a^*$. At a fixed exponent, [theorem 2.3](#) permits only finitely many source exponents, coordinate choices, and products of coefficients of a ; hence each coefficient in (3.3) is a finite sum of derivatives of ordinary analytic germs and is again an analytic germ. The same argument gives strong summability of (3.2). The product rule, the multinomial identity and the formal Taylor translation identity then give all the asserted algebraic identities, the rearrangements being finite at each exponent. Note what this proof does *not* do: it never translates an ordinary germ by a nonzero complex number outside its domain. It translates by an infinitesimal, using derivatives at the original centre. $\square$

Proposition 3.4 (Evaluation kernels and finite jets over $\mathcal{A}_n$). *For every $a \in \mathfrak{m}_K^n$,*

$$\ker \text{ev}_a = \mathfrak{m}_a = (z_1 - a_1, \dots, z_n - a_n), \quad (3.4)$$

and for $N \geq 1$ the Taylor map induces

$$\mathcal{A}_n / \mathfrak{m}_a^N \simeq K[X_1, \dots, X_n] / (X_1, \dots, X_n)^N. \quad (3.5)$$

Proof. At $a = 0$ apply ordinary Taylor division to each coefficient germ: if its constant term vanishes, write it as $\sum_i z_i g_i$ using fixed complex-linear coordinate-division maps, and extend those maps coefficientwise. For general N , subtract the Taylor polynomial of degree below N and divide the remainder by the finitely many monomials of degree N ; polynomial representatives give surjectivity. Translation by [theorem 3.3](#) reduces arbitrary a to zero. $\square$

Proposition 3.5 (Faithful interpretation as functions on the monad). *If $F \in \mathcal{A}_n$ vanishes at every point of $\mathfrak{m}_K^n$ then $F = 0$.*

Proof. The case $n = 0$ is immediate. Normalize a nonzero F as $f_0 + E$ and let m be the ordinary vanishing order of f_0 . Choose $c \in \mathbb{C}^n$ at which the degree- m homogeneous part H_m of f_0 is nonzero. If $E \neq 0$ set $e = v_H(E) > 0$ and choose $\eta > 0$ in Γ with $m\eta < e$, which divisibility permits (for instance $\eta = e/(m+1)$); if $E = 0$ take any $\eta > 0$. At $a = ct^\eta$ one has $f_0(a) = H_m(c)t^{m\eta} + (\text{larger valuation})$ while $v(E(a)) \geq e$, so $F(a) \neq 0$. The case $m = 0$ is included. $\square$

Remark 3.6 (Formal power series are different). The Taylor embedding $\mathcal{A}_n \hookrightarrow K[[z]]$ is not an identification of rings of functions. The coefficientwise formal inverse

$$(z - t^\eta)^{-1} = - \sum_{r \geq 0} t^{-(r+1)\eta} z^r$$

lies in $K[[z]]$ but not in $\mathcal{A}_1$, since its combined Hahn support is not well ordered. Accordingly $z - t^\eta$ is a nonunit of $\mathcal{A}_1$, and it actually vanishes at an infinitesimal point. Completing at a point and working on the entire monad are different operations.

3.2 The common-domain ring $\mathcal{R}_n$

Definition 3.7 (Fixed-domain Hahn families and the common-domain germ ring). For an ordinary connected open $U \subset \mathbb{C}^n$ set

$$\mathcal{H}_\Gamma(U) = \left\{ \sum_{\gamma \in S} f_\gamma t^\gamma : f_\gamma \in \mathcal{O}(U), S \subset \Gamma \text{ well ordered} \right\},$$

and define the common-domain germ ring at the ordinary origin by

$$\mathcal{R}_n = \mathcal{R}_n(\Gamma) = \varinjlim_{U \ni 0} \mathcal{H}_\Gamma(U), \quad (3.6)$$

the limit being over connected ordinary neighbourhoods of 0 with the restriction maps. Set $\mathcal{R}_0 = K$. For $c \in U$ and $\varepsilon \in \mathfrak{m}_K^n$ the evaluation of $F = \sum f_\gamma t^\gamma$ is

$$F(c + \varepsilon) = \sum_{\gamma \in \text{supp } F, \alpha \in \mathbb{N}^n} t^\gamma \frac{\partial^\alpha f_\gamma(c)}{\alpha!} \varepsilon^\alpha. \quad (3.7)$$

One may restrict the limit in (3.6) to ordinary polydisks. Restriction is injective by the ordinary identity theorem, and multiplication of leading coefficient germs shows $\mathcal{R}_n$ is a domain. Every element of $\mathcal{R}_n$ evaluates on all of $\mathfrak{m}_K^n$; its elements are functions on the *whole* monad, not germs at one point of a fine topology.

The quantifier order in (3.6) is the entire point. All coefficient functions of one representative have *one* common domain. The larger expression $\mathcal{O}_{\mathbb{C}^n, 0}((t^\Gamma)) = \mathcal{A}_n$ permits independently shrinking coefficient domains and is a different ring.

Proposition 3.8 (Evaluation, support and translation over $\mathcal{R}_n$). *Formula (3.7) is strongly summable; evaluation respects sums, products and coefficientwise partial derivatives, and is injective on $\mathcal{H}_\Gamma(U)$. Composition with an infinitesimal translation, or with a positive-support perturbation of an ordinary holomorphic map, is well defined whenever the ordinary domains match. For $a \in \mathfrak{m}_K^n$, translation $\tau_a F(z) = F(z + a)$ is an automorphism of $\mathcal{R}_n$ with inverse τ_{-a} , and*

$$\mathcal{R}_n / (z_1 - a_1, \dots, z_k - a_k) \simeq \mathcal{R}_{n-k} \quad (0 \leq k \leq n). \quad (3.8)$$

In particular $\text{ev}_a : \mathcal{R}_n \rightarrow K$ is surjective with kernel $\mathfrak{m}_a = (z_1 - a_1, \dots, z_n - a_n)$.

Proof. Put $P = \bigcup_j \text{supp } \varepsilon_j$. All exponents in (3.7) lie in $\text{supp } F + P^*$, and the multiindices of each fixed total degree give finitely many choices, so [theorem 2.3](#) gives local finiteness and the ordinary Taylor product and composition identities may be rearranged coefficientwise. Differentiating coefficient functions introduces no new exponents. At ordinary points $F(c) = \sum f_\gamma(c)t^\gamma$; if all these vanish, then uniqueness of Hahn coefficients makes every f_γ identically zero, which is injectivity. For translation, the Taylor formula has support bounded by $\text{supp } F$ plus the monoid generated by the coordinates of a , and each derivative coefficient remains holomorphic on the same ordinary domain; the binomial identity gives $\tau_a \tau_{-a} = \text{id}$. For (3.8) at $a = 0$, the ordinary identity

$$h(z_1, z') = h(0, z') + z_1 \frac{h(z_1, z') - h(0, z')}{z_1}$$

has a holomorphic quotient on any polydisk; applying it coefficientwise eliminates one coordinate without changing support, and iterating eliminates k . Translation handles general a . $\square$

Lemma 3.9 (Infinitesimal evaluation detects nonzero elements of $\mathcal{R}_n$). *For every nonzero $F \in \mathcal{R}_n$ there is $a \in \mathfrak{m}_K^n$ with $F(a) \neq 0$.*

Proof. Identical in mechanism to [theorem 3.5](#), and we record it separately because it is a statement about a different ring. Normalize $F = t^\lambda(f_0 + E)$ with $f_0 \neq 0$ and $v_H(E) > 0$, let m be the least total degree of a nonzero homogeneous term of f_0 , and choose $c \in \mathbb{C}^n$ at which that homogeneous polynomial is nonzero. If $E \neq 0$ put $e = \min \text{supp } E > 0$ and $\delta = e/(m+1) \in \Gamma_{>0}$, so $m\delta < e$; otherwise take any $\delta > 0$. At $a = t^\delta c$ the leading term of $f_0(a)$ has valuation $m\delta$, while every value contributed by E has valuation at least e and cannot cancel it. $\square$

The nontriviality of Γ is necessary here: when $\Gamma = \{0\}$ the monad has only its origin and cannot detect ordinary analytic germs. This is the same hypothesis whose necessity is made precise in [theorem 5.6](#).

3.3 The two inclusions are strict

Example 3.10 (No shared ordinary neighbourhood: $\mathcal{R}_n \subsetneq \mathcal{A}_n$). Fix $\eta \in \Gamma_{>0}$ and consider

$$F(z) = \sum_{r \geq 1} \frac{t^{r\eta}}{1 - rz}. \quad (3.9)$$

This is an element of $\mathcal{A}_1$ and it is evaluable at every infinitesimal argument. Yet it cannot be represented by a Hahn family of holomorphic functions on any common ordinary disk centred at zero, so it is not an element of $\mathcal{R}_1$ with those coefficients: its coefficient at $r\eta$ has a pole at $1/r$, and an ordinary disk of positive radius contains $1/r$ for all sufficiently large r . Equality of germwise Hahn expressions forces equality of the coefficient germs, and so does equality as functions on the whole monad, by [theorem 3.5](#). Hence no representative with a common domain exists.

Taking Hahn series and passing to germs therefore do not commute in the naive common-radius sense. This is exactly why invoking a theorem stated only for coefficients holomorphic on one fixed polydisk would leave a gap in the radius-free setting, and it is the reason [theorem 1.1](#) is enforced below without exception.

Remark 3.11 ($\mathcal{A}_n \subsetneq \mathcal{F}_n$). The constant-Hahn-support expression $\sum_{r \geq 0} r! z^r$ lies in $\mathcal{F}_1 = \mathbb{C}[[z]]((t^\Gamma))$ but not in $\mathcal{A}_1$, since $\sum r! z^r$ is not a convergent germ. Both inclusions in (1.1) are therefore strict, and the comparison theorem [theorem 11.5](#) has content beyond a change of notation.

3.4 Taylor calculus on the monad

Differentiation acts coefficientwise on all four rings. Translation gives a Taylor expansion at every infinitesimal centre with a useful valuation bound: if $v_H(F) \geq \lambda$, $a, h \in \mathfrak{m}_K^n$ and $v_{\min}(h) = \rho > 0$, then

$$v\left(F(a+h) - \sum_{|\alpha| < N} \frac{\partial^\alpha F(a)}{\alpha!} h^\alpha\right) \geq \lambda + N\rho, \quad (3.10)$$

because translation does not decrease the lower support bound and every remaining monomial contains at least N factors from h . This gives the expected differential and chain rules on the monad, and the estimates persist after enlarging the workspace. Strong summability remains the definition of the series: these estimates do not turn every sequence of partial sums into a convergent sequence.

4 Weierstrass division and preparation, over each ring

Both rings admit division with the support certificate $S + E^*$, and the two proofs use different ordinary inputs. Over $\mathcal{A}_n$ the input is the ordinary *germ-level* division pair, which is available for each coefficient separately and produces no common domain. Over $\mathcal{R}_n$ the input is a pair of *contour* operators attached once to one product neighbourhood, which never changes the domain. Neither proof can be run in the other setting, and the two theorems are therefore stated separately.

4.1 Division over the radius-free ring $\mathcal{A}_n$

Write $z = (x, w)$ with $x = (z_1, \dots, z_{n-1})$ and $w = z_n$. An ordinary germ $f \in R_n$ is *w-regular of order m* if $f(0, w) = cw^m + O(w^{m+1})$ with $c \neq 0$. The classical Weierstrass theorem gives $f = u_0 P_0$ with u_0 a unit and $P_0 = w^m + \sum_{j < m} p_{0j}(x) w^j$, $p_{0j}(0) = 0$; classical division gives *unique* R_{n-1} -linear maps

$$D : R_n \rightarrow R_n, \quad \Pi : R_n \rightarrow R_{n-1}[w]_{< m}, \quad g = f Dg + \Pi g,$$

so that $D(fg) = g$ and $D(r) = 0$ when $\deg_w r < m$. Standard treatments include [\[12\]](#). Hahn analogues occur in [\[4, 5\]](#); the proof below verifies directly the coefficient and support conditions we need.

Theorem 4.1 (Hahn-germ division over $\mathcal{A}_n$). *Let $F = f + E \in \mathcal{A}_n$, where $f \in R_n$ is w-regular of order $m \geq 1$ and $v_H(E) > 0$. Every $G \in \mathcal{A}_n$ has a unique decomposition*

$$G = FQ + R, \quad Q \in \mathcal{A}_n, \quad R \in \mathcal{A}_{n-1}[w]_{< m}. \quad (4.1)$$

Writing $E_+ = \text{supp } E$ and $S = \text{supp } G$, both Q and R have support contained in $S + E_+^*$. Explicitly

$$Q = (\text{id} + DM_E)^{-1} DG, \quad R = \Pi(G - EQ), \quad (4.2)$$

where M_E is multiplication by E and D, Π act coefficientwise.

Proof. Apply D to the desired equation. Since $D(fQ) = Q$ and $D(R) = 0$ it becomes $(\text{id} + DM_E)Q = DG$. The operator DM_E is a sum of positive-support complex-linear operators on R_n , so [theorem 2.5](#) produces the first formula in (4.2) with the stated support bound. That identity says $D(G - EQ) = Q$; applying ordinary division to $G - EQ$ gives $G - EQ = fQ + \Pi(G - EQ)$, which is (4.1). The remainder has the required degree because Π does so coefficientwise. For uniqueness, subtract two decompositions and apply D : the difference of the quotients lies in the kernel of the invertible operator $\text{id} + DM_E$, hence is zero, and then so is the difference of remainders. All produced coefficients are ordinary analytic germs, since a fixed Hahn coefficient involves only finitely many applications of ordinary division. $\square$

Theorem 4.2 (Preparation and finite projection over $\mathcal{A}_n$). *Under the hypotheses of [theorem 4.1](#) there is a unique factorization*

$$F = UP, \quad P = w^m + \sum_{j < m} p_j(x)w^j, \quad (4.3)$$

with $U \in \mathcal{A}_n$ a unit, each $p_j \in \mathcal{A}_{n-1}$ of nonnegative support, $\text{lc}(P) = P_0$ and $\text{lc}(U) = u_0$. The supports of P and U lie in E_+^* . In particular

$$\mathcal{A}_n/(F) \simeq \mathcal{A}_{n-1}[w]/(P) \quad (4.4)$$

is free of rank m over $\mathcal{A}_{n-1}$, with basis $1, w, \dots, w^{m-1}$.

Proof. Divide w^m by F : $w^m = FQ + R$. The leading coefficient of Q is $D(w^m) = u_0^{-1}$, because $w^m = P_0 + (w^m - P_0)$ is ordinary division by P_0 ; hence Q is a unit by [theorem 3.2](#). Put $P = w^m - R$ and $U = Q^{-1}$; their leading coefficients are P_0 and u_0 , the support assertion for Q, R comes from division, and geometric inversion gives it for U . If $F = \tilde{U}\tilde{P}$ is another such factorization then $w^m = F\tilde{U}^{-1} + (w^m - \tilde{P})$ is another division of w^m with remainder of degree below m , so uniqueness in [theorem 4.1](#) forces $\tilde{P} = P$ and $\tilde{U} = U$. Division gives both surjectivity and uniqueness of polynomial representatives in (4.4). $\square$

For a general nonzero $F \in \mathcal{A}_n$, first factor out $t^{v_H(F)}$; then a complex-linear change of coordinates makes the leading germ w -regular, by choosing a direction on which the first nonzero homogeneous part does not vanish. Such a change acts coefficientwise and preserves all exponent supports. These operations reduce any nonunit to the situation of [theorem 4.2](#) with $m \geq 1$.

Corollary 4.3 (Fibres of a prepared hypersurface over $\mathcal{A}_n$). *For $a' \in \mathfrak{m}_K^{n-1}$ the fibre algebra of (4.4) is $K[w]/P(a', w)$. It has length m , all its roots are infinitesimal, and it has m distinct roots exactly when the discriminant of $P(a', w)$ is nonzero.*

Proof. The lower coefficients of $P(a', w)$ lie in $\mathfrak{m}_K$: their exponent-zero coefficient germs vanish at the origin in the x variables, and all remaining Hahn terms have positive valuation. A root of negative valuation would make w^m the unique term of least valuation; a root of valuation zero would reduce to a nonzero root of w^m over $\mathbb{C}$. Both are impossible. Algebraic closedness of K and the discriminant criterion finish the proof. $\square$

This is a genuine finite-mapping statement with infinitesimal parameter values, not merely a one-variable factorization at a single point. It does not assert that the roots can be globally labelled by single-valued analytic functions through a branch locus.

4.2 Division over the common-domain ring $\mathcal{R}_n$

The ordinary input here is different, and it is what keeps the domain fixed. Write $z = (x, w)$ with $x \in \mathbb{C}^{n-1}$, let V be an ordinary polydisk and $D_R = \{|w| < R\}$, and let $P_0(x, w)$ be monic of degree d in w with coefficients holomorphic on V and every root of $P_0(x, \cdot)$ in D_r with $r < R$ fixed. Choose $r < \rho < R$ and set, for $h \in \mathcal{O}(V \times D_R)$,

$$R_0 h(x, w) = \frac{1}{2\pi i} \int_{|\zeta|=\rho} \frac{h(x, \zeta)}{P_0(x, \zeta)} \cdot \frac{P_0(x, \zeta) - P_0(x, w)}{\zeta - w} d\zeta, \quad D_0 h = \frac{h - R_0 h}{P_0}. \quad (4.5)$$

The quotient involving P_0 is a polynomial in w of degree below d , so $R_0 h$ has that degree with coefficients holomorphic on V ; for $|w| < \rho$ the Cauchy formula shows $h - R_0 h$ is divisible by P_0 , and since all zeros of P_0 lie in the smaller disk the quotient extends holomorphically to all of $V \times D_R$. Thus

$$h = R_0 h + P_0 D_0 h, \quad \deg_w R_0 h < d, \quad (4.6)$$

uniquely, and both operators are complex-linear endomorphisms of the same space $\mathcal{O}(V \times D_R)$. That last clause is the resource unavailable over $\mathcal{A}_n$.

Theorem 4.4 (Parameterized Hahn preparation over $\mathcal{R}_n$). *Let $F \in \mathcal{R}_n$ be nonzero, normalized as $F = t^\lambda(f_0 + E)$ with $v_H(E) > 0$, and suppose $f_0(0, w)$ has a zero of order $d \geq 1$ at $w = 0$. Choose an ordinary product neighbourhood on which ordinary preparation gives $f_0 = u_0 P_0$ with u_0 nowhere zero, P_0 monic of degree d and $P_0(0, w) = w^d$. On a possibly smaller product neighbourhood there are unique normalized factors*

$$F = t^\lambda u_0 (P_0 + A)(1 + B), \quad \deg_w A < d, \quad (4.7)$$

with A and B of strictly positive Hahn support. If $E' = E/u_0$ has support E_+ , then $\text{supp } A, \text{supp } B \subseteq E_+^* \setminus \{0\}$. Every coefficient function of every factor is holomorphic on this one neighbourhood.

Proof. Ordinary Weierstrass preparation is used once, on f_0 [12]. Shrink the parameter polydisk so that the roots of P_0 stay in a fixed smaller w -disk, giving the operators (4.5). Dividing by $t^\lambda u_0$, the equation to solve is $E' = A + P_0 B + AB$. Proceed by well-founded recursion on $E_+^* \setminus \{0\}$: at exponent ν put

$$h_\nu = E'_\nu - \sum_{\substack{\alpha+\beta=\nu \\ \alpha, \beta > 0}} A_\alpha B_\beta, \quad A_\nu = R_0 h_\nu, \quad B_\nu = D_0 h_\nu. \quad (4.8)$$

The displayed sum is finite by theorem 2.3 and uses only smaller exponents, and (4.6) gives the desired equation at ν . The support lemma proves the assembled factors exist. Crucially R_0, D_0 never change the chosen ordinary domain. For uniqueness, compare two normalized pairs at the least exponent where either differs: the product AB cannot contribute a first difference there, since both factors have positive support, so $\Delta A_\nu + P_0 \Delta B_\nu = 0$ and ordinary division uniqueness makes both differences zero. The factor $1 + B$ is a unit by positive-support geometric inversion and u_0 is an ordinary unit. $\square$

Theorem 4.5 (Parameterized Hahn division over $\mathcal{R}_n$). *Let $P = P_0 + A$ be the monic factor of theorem 4.4 and let G be a Hahn-holomorphic datum on its product neighbourhood. There are unique Q and R on that same neighbourhood with*

$$G = PQ + R, \quad \deg_w R < d, \quad (4.9)$$

and with $E_A = \text{supp } A \subset \Gamma_{>0}$ both outputs have support in $\text{supp } G + E_A^*$. They are given by

$$Q = \sum_{k \geq 0} (-D_0(A \cdot))^k D_0 G, \quad R = R_0(G - AQ). \quad (4.10)$$

For an arbitrary germ $G \in \mathcal{R}_n$ one first chooses a common smaller product neighbourhood; no subsequent shrinking is needed.

Proof. Apply D_0 to (4.9): the equation for Q is $Q + D_0(AQ) = D_0 G$, and [theorem 2.5](#) gives (4.10) with its support bound. Set $R = R_0(G - AQ)$ and use (4.6) to recover (4.9). If $PQ + R = 0$ with $\deg_w R < d$, the same invertible operator gives $Q = 0$ and then $R = 0$. $\square$

Corollary 4.6 (Finite free hypersurface quotient over $\mathcal{R}_n$). *For P as above, $\mathcal{R}_n/(P) \simeq \bigoplus_{j=0}^{d-1} \mathcal{R}_{n-1} w^j$ as an $\mathcal{R}_{n-1}$ -module, and for each $a' \in \mathfrak{m}_K^{n-1}$ all d roots of $P(a', w)$ are infinitesimal, counted with multiplicity.*

Proof. Division gives the module isomorphism. The lower coefficients of $P_0(a', w)$ are infinitesimal because their ordinary values at $a' = 0$ vanish, and the coefficients of $A(a', w)$ are infinitesimal as well. If a root w were not infinitesimal, dividing the equation by w^d would give 1 plus an infinitesimal equal to zero. Algebraic closedness of K supplies all d roots. $\square$

Remark 4.7 (Why both division theorems are kept). [Theorem 4.1](#) applies to coefficient families with radii tending to zero, which [theorem 4.5](#) cannot reach; the element (3.9) is such a family. [Theorem 4.5](#) delivers its quotient and remainder on *one named ordinary domain*, which [theorem 4.1](#) cannot deliver, and that fixed domain is what later permits Cartan's theorem B ([theorem 9.5](#)) and a fixed ordinary residue cycle ([theorem 10.1](#)). Neither theorem implies the other.

5 Noetherian geometry and the Nullstellensatz over $\mathcal{A}_n$

Everything in this section is a statement about the radius-free ring $\mathcal{A}_n = \mathbb{C}\{z\}((t^\Gamma))$ of [theorem 3.1](#). The corresponding statements for $\mathcal{R}_n$ are in [section 6](#) and are not consequences of these.

5.1 Finite generation

Theorem 5.1 (Noetherianity of $\mathcal{A}_n$). *For every $n \geq 0$ the ring $\mathcal{A}_n$ is Noetherian.*

Proof. Induct on n , starting from the field $\mathcal{A}_0 = K$. Let $I \subseteq \mathcal{A}_n$ be a nonzero proper ideal and choose $0 \neq F \in I$. Normalize and change complex-linear coordinates so that [theorem 4.2](#) applies with degree $m \geq 1$. Then $\mathcal{A}_n/(F)$ is finite free over $\mathcal{A}_{n-1}$, hence a Noetherian $\mathcal{A}_{n-1}$ -module by the induction hypothesis, so its submodule $I/(F)$ is finitely generated over $\mathcal{A}_{n-1}$. Lifting those generators and adjoining F gives finitely many ideal generators of I . The zero and unit ideals need no argument. $\square$

5.2 Every maximal ideal of $\mathcal{A}_n$ is an evaluation ideal

Theorem 5.2 (Weak Nullstellensatz over $\mathcal{A}_n$). *Every maximal ideal of $\mathcal{A}_n$ is $\mathfrak{m}_a$ for a unique $a \in \mathfrak{m}_K^n$, and its residue field is K .*

Proof. The case $n = 0$ is clear. Assume the statement in dimension $n - 1$ and let M be maximal in $\mathcal{A}_n$. Since $\mathcal{A}_n$ is not a field for $n \geq 1$, M contains a nonzero element; make it regular as above, obtaining the finite integral algebra $\mathcal{A}_n/(F) \simeq \mathcal{A}_{n-1}[w]/(P)$. The contraction of $M/(F)$ to $\mathcal{A}_{n-1}$ is maximal by integrality, hence by induction is evaluation at some $a' \in \mathfrak{m}_K^{n-1}$ with residue field K ; therefore $\mathcal{A}_n/M$ is a finite field extension of K and equals K . Let a_n be the image of w in that field; it satisfies $P(a', a_n) = 0$, so [theorem 4.3](#) gives $a_n \in \mathfrak{m}_K$. The coordinate images thus form an infinitesimal tuple a ; the ideal $(z - a)$ is contained in M and is maximal by [theorem 3.4](#), so $M = (z - a)$. Reversing the complex-linear coordinate change keeps the coordinates infinitesimal, and evaluating the coordinate functions gives uniqueness. $\square$

This proof never assumes that an abstract field-valued homomorphism commutes with an infinite Hahn sum. The finite integral quotient identifies the coordinate images, and the independently proved kernel formula [\(3.4\)](#) identifies the ideal. $\square$

5.3 The strong theorem

For $X \subseteq \mathfrak{m}_K^n$ put $I(X) = \{F \in \mathcal{A}_n : F(a) = 0 \ \forall a \in X\}$, and for an ideal I let $Z(I) \subseteq \mathfrak{m}_K^n$ be its common zero set.

Theorem 5.3 (Jacobson property and strong Nullstellensatz over $\mathcal{A}_n$). *The ring $\mathcal{A}_n$ is Jacobson, and for every ideal $I \subseteq \mathcal{A}_n$*

$$I(Z(I)) = \sqrt{I}. \quad (5.1)$$

Proof. We prove the Jacobson assertion by induction on n , the base being a field. Let $\mathfrak{p} \neq 0$ be prime, choose $0 \neq F \in \mathfrak{p}$ and prepare it after a coordinate change; then $\mathcal{A}_n/\mathfrak{p}$ is finite over $\mathcal{A}_{n-1}/(\mathfrak{p} \cap \mathcal{A}_{n-1})$, which is Jacobson by induction and passage to a quotient, and a finite algebra over a Jacobson ring is Jacobson [[19](#), §10.35]. Hence $\mathfrak{p}$ is the intersection of the maximal ideals containing it.

The zero prime requires a separate argument; the finite-quotient reduction does not apply to it. By [theorem 3.5](#) the intersection of all evaluation ideals $\mathfrak{m}_a$ is zero. Thus every prime is an intersection of maximal ideals, which is the Jacobson property, and by [theorem 5.2](#) those maximal ideals are precisely the evaluation ideals. Intersecting the maximal ideals containing I gives [\(5.1\)](#). $\square$

A family of Hahn-germ equations with no common infinitesimal zero therefore generates the unit ideal, and because $\mathcal{A}_n$ is Noetherian the assertion has a finite witness: finitely many F_j and $G_j \in \mathcal{A}_n$ with $\sum_j G_j F_j = 1$. The theorem gives existence, not a uniform bound on the complexity of finding such a witness from arbitrary support presentations.

5.4 Regularity, completion, and two warnings

Theorem 5.4 (Local geometry of $\mathcal{A}_n$). *The ring $\mathcal{A}_n$ has dimension n and is regular. For every $a \in \mathfrak{m}_K^n$,*

$$\widehat{(\mathcal{A}_n)_{\mathfrak{m}_a}} \simeq K[[X_1, \dots, X_n]], \quad X_i = z_i - a_i. \quad (5.2)$$

Proof. Translate to $a = 0$. For $0 \leq r \leq n$, coefficientwise restriction gives $\mathcal{A}_n/(z_1, \dots, z_r) \simeq \mathcal{A}_{n-r}$, a domain, so the chain of coordinate primes remains strict after localizing at $\mathfrak{m}_0$ and that local ring has dimension at least n . Its maximal ideal is generated by the n coordinate functions, so the standard Noetherian local dimension bound gives dimension at most n , and the finite-jet calculation

(3.5) shows the cotangent space has dimension n ; hence the local ring is regular. By theorem 5.2 every maximal localization is of this form, so $\mathcal{A}_n$ has dimension n and all localizations at primes are regular, regularity being preserved under localization. Finally an element outside $\mathfrak{m}_a$ is already invertible modulo each $\mathfrak{m}_a^N$, since its constant term there is nonzero, so localization does not change the finite-jet quotients (3.5); taking the inverse limit gives (5.2). $\square$

Remark 5.5 ($\mathcal{A}_n$ is not local). The ring $\mathcal{A}_n$ simultaneously records every infinitesimal point, so it has one maximal ideal for each $a \in \mathfrak{m}_K^n$; its ordinary coefficient ring R_n is local by contrast. This is exactly why an infinitesimal Nullstellensatz can coexist with nontrivial local analytic geometry, and it is the formal content of the distinction between a stalk on the whole monad and a fine germ at one surcomplex point. Nothing below identifies the two.

Remark 5.6 (The nonzero-value-group hypothesis is essential). If $\Gamma = \{0\}$ then $\mathfrak{m}_K^n = \{0\}$ and $\mathcal{A}_n = R_n$. For $n > 0$ the vanishing ideal of that single point is (z) , whereas $\sqrt{(0)} = (0)$; thus (5.1) fails at $I = (0)$. The hypothesis is used essentially in theorem 3.5. It is not an aesthetic restriction.

6 Noetherian geometry, normalization and finite maps over $\mathcal{R}_n$

Everything in this section is a statement about the common-domain ring $\mathcal{R}_n$ of theorem 3.7. The statements parallel those of section 5 but are proved by a different route, and the normalization and finite-map results of section 7 have no counterpart there.

Theorem 6.1 (Noetherian monad ring $\mathcal{R}_n$). *The ring $\mathcal{R}_n$ is a Noetherian domain of Krull dimension n .*

Proof. The domain assertion follows from multiplication of leading coefficient germs. For Noetherianity induct on n from $\mathcal{R}_0 = K$. Let $I \subset \mathcal{R}_n$ be nonzero and choose $0 \neq F \in I$. An ordinary invertible linear change of coordinates makes its leading ordinary coefficient regular in the last variable, by choosing a direction where its first nonzero homogeneous Taylor polynomial does not vanish. If that coefficient has order zero then F is a unit after shrinking the ordinary neighbourhood and $I = \mathcal{R}_n$. Otherwise theorem 4.4 replaces F , up to a unit, by a monic P of positive degree in the last variable, and by theorem 4.6 the quotient $\mathcal{R}_n/(F)$ is finite free over $\mathcal{R}_{n-1}$; as that ring is Noetherian, $I/(F)$ is finitely generated, and lifting generators and adjoining F generates I .

For the dimension, the coordinate ideals $(0) \subsetneq (z_1) \subsetneq \cdots \subsetneq (z_1, \dots, z_n)$ are prime by (3.8), giving the lower bound n . For the upper bound, prepend (0) to any prime chain if necessary and choose a nonzero element of its first nonzero prime; preparation makes the quotient by that element finite integral over $\mathcal{R}_{n-1}$, and integral extensions preserve dimension, so the remaining chain has length at most $n - 1$. The elementary integral-extension facts used here — incomparability, going up and lying over — are the standard results recorded in [19]. $\square$

Proposition 6.2 (Noether normalization over $\mathcal{R}_n$). *Let $\mathfrak{p} \subset \mathcal{R}_n$ be prime and $d = \dim(\mathcal{R}_n/\mathfrak{p})$. After an ordinary linear coordinate change, the first d coordinates induce a finite injective homomorphism*

$$\mathcal{R}_d \hookrightarrow A = \mathcal{R}_n/\mathfrak{p}. \quad (6.1)$$

Proof. Induct on n . If $\mathfrak{p} = 0$ take $d = n$. Otherwise prepare a nonzero element of $\mathfrak{p}$ in the last variable; then A is finite over $\mathcal{R}_{n-1}/\mathfrak{q}$ with $\mathfrak{q}$ the contraction of $\mathfrak{p}$, and by induction that domain is finite over an injective coordinate copy of some $\mathcal{R}_d$. Finiteness is transitive. All coordinate changes in the induction act ordinarily on the remaining coordinates, so their composition is an ordinary

linear change on $\mathbb{C}^n$; dimension is preserved in each finite integral extension, which identifies the final number of coordinates with $\dim A$. $\square$

This proposition has no counterpart over $\mathcal{A}_n$ in this report, and it is the input to both the strong Nullstellensatz proof below and the finite-map theorem of [section 7](#).

Theorem 6.3 (Weak monad Nullstellensatz over $\mathcal{R}_n$). *Every maximal ideal of $\mathcal{R}_n$ is uniquely $\mathfrak{m}_a = (z_1 - a_1, \dots, z_n - a_n)$ with $a \in \mathfrak{m}_K^n$; its residue field is K and its residue map is Hahn evaluation at a .*

Proof. Induct on n , the case $n = 0$ being immediate. For $n > 0$, $\mathcal{R}_n$ is not a field since evaluation at zero shows z_1 is a nonunit, so a maximal ideal $\mathfrak{m}$ is nonzero. Prepare an element of $\mathfrak{m}$ and use the resulting finite extension of $\mathcal{R}_{n-1}$; the contraction $\mathfrak{n}$ is maximal, since a subring over which a field is integral is a field, and by induction $\mathcal{R}_{n-1}/\mathfrak{n} = K$. Hence $\mathcal{R}_n/\mathfrak{m}$ is a finite extension of the algebraically closed K , so equals K .

Let $a_j \in K$ be the image of z_j . If $v(a_j) < 0$ then $z_j - a_j$ has a nonzero constant as leading Hahn coefficient after normalization and is a unit. If $v(a_j) = 0$ its leading coefficient is $z_j - c$ with $c \in \mathbb{C}^\times$, a unit on a sufficiently small ordinary neighbourhood of zero. Both cases contradict $z_j - a_j \in \mathfrak{m}$, so every a_j is infinitesimal or zero. By (3.8) the ideal generated by these coordinate differences already has quotient K and is precisely the evaluation kernel, so it equals $\mathfrak{m}$. The coordinate images give uniqueness. $\square$

Theorem 6.4 (Strong monad Nullstellensatz over $\mathcal{R}_n$). *For an ideal $I \subset \mathcal{R}_n$ put $V(I) = \{a \in \mathfrak{m}_K^n : F(a) = 0 \text{ for all } F \in I\}$. Then*

$$I(V(I)) = \sqrt{I}, \quad (6.2)$$

equivalently $\mathcal{R}_n$ is Jacobson and all its closed points are its infinitesimal evaluation points.

Proof. We first show a prime $\mathfrak{p}$ is the intersection of the maximal ideals containing it. Let $A = \mathcal{R}_n/\mathfrak{p}$ and $0 \neq g \in A$, and normalize A over $B = \mathcal{R}_d$ as in [theorem 6.2](#). The finite-dimensional domain $A \otimes_B \text{Frac}(B)$ is a field, so g^{-1} lies in it; clearing a denominator gives $gh = b$ with $h \in A$ and $0 \neq b \in B$. By [theorem 3.9](#) choose an evaluation maximal ideal $\mathfrak{n}$ of B not containing b ; lying over for the integral extension $B \subset A$ gives a maximal ideal M of A over $\mathfrak{n}$, and $gh = b$ implies $g \notin M$. Pulling M back to $\mathcal{R}_n$ and applying [theorem 6.3](#) produces a point of $V(\mathfrak{p})$ where any representative of g is nonzero.

If $F \notin \sqrt{I}$, choose a prime $\mathfrak{p} \supseteq I$ not containing F ; the preceding argument finds an evaluation maximal ideal containing $\mathfrak{p}$ but not F , so F does not vanish on all of $V(I)$. The reverse inclusion in (6.2) is immediate in the field K . $\square$

Remark 6.5 (The two Jacobson proofs are genuinely different). [Theorem 5.3](#) obtains the Jacobson property of $\mathcal{A}_n$ by induction through finite algebras over Jacobson rings, and then handles the zero prime separately by [theorem 3.5](#), because the finite-quotient reduction does not apply to it. [Theorem 6.4](#) instead runs Noether normalization, inverts g in the generic fibre, and uses lying over together with [theorem 3.9](#) to produce the required point; the zero prime needs no separate treatment because normalization applies to it. Neither argument is a specialization of the other, and neither theorem may be transferred between the rings.

Proposition 6.6 (Local coordinates and completion over $\mathcal{R}_n$). *For $a \in \mathfrak{m}_K^n$ the local ring $(\mathcal{R}_n)_{\mathfrak{m}_a}$ is regular of dimension n , and*

$$\widehat{(\mathcal{R}_n)_{\mathfrak{m}_a}} \simeq K[[h_1, \dots, h_n]], \quad h_j = z_j - a_j. \quad (6.3)$$

The image of F in this completion is its formal Taylor series at a .

Proof. Translate a to zero. The maximal ideal is generated by the n coordinate functions and the coordinate prime chain survives localization, so the local dimension is at least n , hence exactly n by [theorem 6.1](#); a Noetherian local ring of dimension n whose maximal ideal has n generators is regular. For every $r \geq 1$, ordinary Taylor division applied coefficientwise gives a unique decomposition into the polynomial of total degree below r and a member of $(z_1, \dots, z_n)^r$, whence $\mathcal{R}_n/\mathfrak{m}_0^r \simeq K[z]/(z)^r$. Localizing does not change this quotient, whose radical is already maximal, and the inverse limit over r gives (6.3). Translation and [theorem 3.8](#) identify the coefficients with the Taylor coefficients at a . $\square$

Consequently, for an isolated zero of $F_1, \dots, F_n$ at a the local intersection multiplicity is unambiguously

$$e_a = \dim_K \frac{K[[h_1, \dots, h_n]]}{(F_1(a+h), \dots, F_n(a+h))} \quad (6.4)$$

when finite, and it equals the length of the corresponding local quotient of $\mathcal{R}_n$, since completion preserves finite-length modules. The same formula holds over $\mathcal{A}_n$ by [theorem 5.4](#); it is one formula instantiated twice, not one theorem transferred.

Remark 6.7 (Neither ring is local). [Theorem 6.4](#) is not the ordinary complex local Nullstellensatz restated with a different scalar field: $\mathcal{R}_n$ has many maximal ideals, whereas $\mathbb{C}\{z\}$ is local. The same remark applies to $\mathcal{A}_n$ ([theorem 5.5](#)). This is the sense in which these rings record an entire infinitesimal monad rather than a single fine germ.

7 Finite maps of monad zero sets over $\mathcal{R}_n$

Theorem 7.1 (Finite projection and generic fibres over $\mathcal{R}_n$). *Let $\mathfrak{p} \subset \mathcal{R}_n$ be prime and let $A = \mathcal{R}_n/\mathfrak{p}$ have dimension d . After an ordinary linear coordinate change the coordinate projection*

$$\pi : V(\mathfrak{p}) \longrightarrow \mathfrak{m}_K^d$$

is surjective with finite fibres, induced by a finite injective map $B = \mathcal{R}_d \hookrightarrow A$. If A is generated by N elements as a B -module, every fibre has scheme length at most N . Let $r = [\text{Frac}(A) : \text{Frac}(B)]$. There is a nonzero $s \in B$ such that every $b \in \mathfrak{m}_K^d$ with $s(b) \neq 0$ has exactly r distinct preimages, each reduced; such b exist.

Proof. Use [theorem 6.2](#). Lying over makes the map on maximal ideals surjective, and by [theorem 6.3](#) these are exactly the evaluation points on both sides, which gives surjectivity on monads. For an evaluation maximal ideal $\mathfrak{n}_b \subset B$ the fibre algebra $A/\mathfrak{n}_b A$ is a nonzero finite-dimensional K -algebra spanned by the N module generators, giving the length bound and finiteness.

Set $L = \text{Frac}(A)$, $L_0 = \text{Frac}(B)$. The extension L/L_0 is finite and separable since the characteristic is zero, so by the primitive element theorem and clearing denominators there is a primitive element θ in a localization of A . The finitely many module generators of A are L_0 -linear combinations of $1, \theta, \dots, \theta^{r-1}$. Localize at one nonzero $s \in B$ so that these expressions, the coefficients of the monic minimal polynomial $q(T)$ of θ , and θ itself are defined over B_s , and enlarge s to make the nonzero discriminant of q invertible. Then

$$A_s \simeq B_s[T]/(q), \quad \deg q = r, \quad \text{disc}(q) \in B_s^\times :$$

the expressions for the generators give surjectivity, and monic division together with independence of $1, \theta, \dots, \theta^{r-1}$ over L_0 gives injectivity. Each such fibre is therefore the algebra of a degree- r polynomial with nonzero discriminant over the algebraically closed K , so it has r distinct points. Finally [theorem 3.9](#) supplies a b avoiding $s = 0$. $\square$

The word *finite* here has its algebraic meaning throughout: finite module and finite fibres. No claim of properness, compactness, or classical path lifting in the full surreal fine topology is included. Two senses of finiteness occur in this report and both are purely algebraic and free-of-finite-rank, never metric: the *finite projection* of [theorem 4.2](#) and [theorem 4.6](#), in which $\mathcal{A}_n/(F)$ respectively $\mathcal{R}_n/(P)$ is free of rank m over the ring in one fewer variable with basis $1, w, \dots, w^{m-1}$; and the *finite fibre* of [theorem 4.3](#) and [theorem 7.1](#), an algebra of length m over K all of whose roots are infinitesimal.

Remark 7.2 (What $\mathcal{A}_n$ is not given here). We do not assert a Noether normalization or a finite-projection theorem for $\mathcal{A}_n$. The normalization proof above uses preparation with holomorphic parameters on one fixed neighbourhood, and we have not verified that the germ-level preparation of [theorem 4.2](#) supports the same induction. The hypersurface case [theorem 4.3](#) is available over $\mathcal{A}_n$, and nothing more is claimed.

8 Isolated complete intersections: the classical ingredients

The deformation package — division with the $S + E^*$ certificate, conservation of μ , characters as actual infinitesimal evaluations, the perfect residue pairing and the trace–Jacobian identity — is common property of this subject and is stated at its strongest hypotheses in the companion report on finite deformations [\[10\]](#). What is distinctive here, and what this report contributes, is that the package survives the *removal of the common-radius hypothesis*: it holds over the strictly larger ring $\mathcal{A}_n$, which by [theorem 3.10](#) contains elements no common-domain theorem can reach. We therefore state the theorem over each ring, give the two proofs, and refer to [\[10\]](#) for the parts of the package that are neither strengthened nor re-proved here.

Throughout [sections 8 to 11](#), let $f = (f_1, \dots, f_n) \in R_n^n$ vanish at the origin with

$$0 < \mu = \dim_{\mathbb{C}} R_n/(f_1, \dots, f_n) < \infty. \quad (8.1)$$

This is the algebraic form of an isolated common zero. Since R_n is regular local and the ideal is generated by n elements with zero-dimensional quotient, it is a complete intersection. Choose monomials $b_1, \dots, b_\mu$ whose classes form a $\mathbb{C}$ -basis of $B_0 = R_n/(f)$; such a choice exists because some power of (z) lies in (f) , so finitely many monomials span the quotient and a subset is a basis.

8.1 A linear division splitting

Let $i : \mathbb{C}^\mu \rightarrow R_n$ send coordinate vectors to the chosen monomials and let $p : R_n \rightarrow \mathbb{C}^\mu$ be the coordinate map of the quotient in this basis. There is a complex-linear $h : R_n \rightarrow R_n^n$ with

$$g = i p(g) + \sum_{j=1}^n f_j h_j(g), \quad p i = \text{id}, \quad h i = 0 : \quad (8.2)$$

take a complex-linear right inverse of the surjection $R_n^n \rightarrow (f)$ and apply it to $g - i p(g)$. This is a choice of linear splitting, not a claim of uniqueness of the quotients $h_j(g)$; one may instead use a

fixed constructive local division procedure when an effective presentation is available. No ordinary norm bound on this splitting is assumed. It maps analytic germs to analytic germs, which is exactly what coefficientwise Hahn extension requires, and this freedom is what removes the need to choose one analytic neighbourhood for infinitely many input coefficients.

8.2 Local residues and duality, imported

We explicitly import the following classical facts; they are not novelty claims. For an isolated tuple $g = (g_1, \dots, g_n)$ in R_n the ordinary local Grothendieck residue is the complex-linear functional

$$\text{res}_g(q) = \frac{1}{(2\pi i)^n} \int_{\{|g_j|=\epsilon_j\}} \frac{q(z) dz_1 \wedge \dots \wedge dz_n}{g_1 \cdots g_n}, \quad (8.3)$$

with sufficiently small regular radii and the orientation given by increasing arguments of the g_j . This is an *ordinary* complex-valued contour integral over an ordinary real cycle in $\mathbb{C}^n$; below it is extended coefficientwise to Hahn numerators, and this report never constructs a surcomplex contour. For background and the equivalent algebraic treatment see [20, 18, 3, 13].

The residue annihilates (g) and induces a nondegenerate bilinear pairing on $R_n/(g)$. For coordinate powers,

$$\text{res}_{(z_1^{d_1}, \dots, z_n^{d_n})}(q) = [z_1^{d_1-1} \cdots z_n^{d_n-1}] q. \quad (8.4)$$

For $k \in \mathbb{N}^n$ write $f^{k+1} = (f_1^{k_1+1}, \dots, f_n^{k_n+1})$; cancellation of a denominator power gives

$$\text{res}_{f^{k+1}}(f_j q) = \begin{cases} 0, & k_j = 0, \\ \text{res}_{f^{k+1-e_j}}(q), & k_j > 0, \end{cases} \quad (8.5)$$

the denominator tuple in the second line still having every exponent positive. These are special cases of the residue transformation rule. We also use the ordinary finite-family trace identity: in a sufficiently small analytic deformation g_u of f by finitely many ordinary parameters u , the cluster originating at zero has a finite free algebra of rank μ over the parameter-germ ring and

$$\text{res}_{g_u}^{\text{cluster}}(q J_{g_u}) = \text{Tr}(M_q), \quad (8.6)$$

the cluster residue being the sum over all points in a fixed small representative neighbourhood, multiple points included. Its algebraic form is the trace-residue theorem of [18]; deformation and computational formulations appear in [15, 3]. A full justification of the finite-family input and its exact role is in [section A](#).

The classical facts above are used on *one finite family at a time*. We never assert that all coefficient germs of a Hahn perturbation admit a common contour or a common analytic representative. Over $\mathcal{R}_n$ a common representative does exist, by definition of the ring, and [theorem 10.1](#) exploits it; over $\mathcal{A}_n$ it does not, and [theorem 10.4](#) is the substitute.

8.3 The perturbation residue and its normalization

Keep (8.1) and put

$$F_j = f_j + E_j, \quad v_H(E_j) > 0, \quad E = \bigcup_j \text{supp } E_j \subset \Gamma_{>0}, \quad (8.7)$$

with E_j in $\mathcal{A}_n$ or in $\mathcal{R}_n$ as the context specifies. Extending each ordinary residue functional coefficientwise to Hahn numerators, define for H in the ambient ring

$$\boxed{\operatorname{Res}_F(H) = \sum_{k \in \mathbb{N}^n} (-1)^{|k|} \operatorname{res}_{f^{k+1}}(H E_1^{k_1} \cdots E_n^{k_n}).} \quad (8.8)$$

The sign $(-1)^{|k|}$, together with the factor $1/(2\pi i)^n$ on the contour side of (8.3), is the normalization, and it is chosen so that $\operatorname{Res}_F(J_F) = \mu$ and, at simple zeros, $\operatorname{Res}_F(H) = \sum_a H(a)/J_F(a)$.

We state emphatically, once: (8.8) is *motivated* by expanding each reciprocal $(f_j + E_j)^{-1}$, but the displayed series, not a putative surcomplex contour integral, *is* the definition. Supplying an actual surcomplex contour that represents it is a separate task, addressed in the companion report on contours [11]. The functional Λ_F of section 10.1 is not a counterexample to this: it expands an *ordinary* cycle coefficientwise, and it is defined only over $\mathcal{R}_n$, where such a cycle exists.

Three senses of multiplicity are kept apart throughout and should not be conflated:

1. the *global colength* $\mu = \dim_{\mathbb{C}} R_n/(f) = \dim_K B_F$, the total length, conserved;
2. the *local algebraic multiplicity* $e_a = \dim_K B_a$ in the Artin decomposition $B_F \simeq \prod_{a \in Z_F} B_a$, with $e_a \geq 1$ and $\sum_a e_a = \mu$;
3. the same e_a computed *formally at the actual displaced zero* a rather than at its standard part, by (6.4).

9 The deformation theorem over each ring

9.1 The radius-free form, over $\mathcal{A}_n$

Lemma 9.1 (Well-definedness and annihilation over $\mathcal{A}_n$). *For $E_j \in \mathcal{A}_n$, the sum (8.8) is strongly summable and defines a K -linear functional $\mathcal{A}_n \rightarrow K$. If $S = \operatorname{supp} H$ then $\operatorname{supp} \operatorname{Res}_F(H) \subseteq S + E^*$. For every j and H , $\operatorname{Res}_F(F_j H) = 0$.*

Proof. The ordinary residue operators do not enlarge Hahn support. A term of multi-index k is supported on sums of a source exponent from S and $|k|$ positive perturbation exponents, so theorem 2.3 proves both well-ordering and finiteness of the full family at each exponent, even though the residue functional itself varies with k ; this also gives K -linearity by the permitted finite rearrangements.

For annihilation, split $F_j H = f_j H + E_j H$. In the first part, terms with $k_j = 0$ vanish by (8.5); in the remaining terms cancel one power of f_j and replace k by $k - e_j$. The result is the negative of the second part of the sum, the reindexing being legitimate coefficientwise by the finiteness already proved. Hence $\operatorname{Res}_F(F_j H) = 0$. $\square$

Extend i, p, h from (8.2) coefficientwise and define the positive-support operator

$$T : G \mapsto \sum_{j=1}^n E_j h_j(G). \quad (9.1)$$

For $G \in \mathcal{A}_n$ set

$$L_G = (\operatorname{id} + T)^{-1} G, \quad \operatorname{NF}_F(G) = p(L_G) \in K^\mu, \quad Q_j(G) = h_j(L_G), \quad (9.2)$$

the inverse being the strong Neumann sum of theorem 2.5.

Lemma 9.2 (Spanning with a support certificate, over $\mathcal{A}_n$). *For every $G \in \mathcal{A}_n$,*

$$G = \sum_{r=1}^{\mu} \text{NF}_F(G)_r b_r + \sum_{j=1}^n F_j Q_j(G), \quad (9.3)$$

and the supports of all displayed coordinates and quotients lie in $\text{supp } G + E^$.*

Proof. The support claim follows from [theorem 2.5](#) and the coefficientwise action of p, h . Since $G = L_G + TL_G$, applying the ordinary splitting (8.2) to L_G gives $G = ip(L_G) + \sum_j f_j h_j(L_G) + \sum_j E_j h_j(L_G)$, which is (9.3). $\square$

This proves that the b_r span the perturbed quotient. Spanning alone is not conservation of multiplicity: the dimension might still have dropped. The residue pairing is what prevents that drop.

Theorem 9.3 (Finite complete-intersection deformation over $\mathcal{A}_n$: the radius-free strengthening). *Assume (8.1) and (8.7) with $E_j \in \mathcal{A}_n$. Then the classes of $b_1, \dots, b_\mu$ form a K -basis of*

$$B_F = \mathcal{A}_n / (F_1, \dots, F_n);$$

in particular $\dim_K B_F = \mu$. The functional Res_F descends to B_F and the pairing

$$B_F \times B_F \longrightarrow K, \quad (u, v) \longmapsto \text{Res}_F(uv) \quad (9.4)$$

is perfect. For the fixed basis, (9.2) is the unique normal-coordinate map, independently of the auxiliary splitting h .

Proof. By [theorem 9.1](#) the residue descends to the quotient. Its Gram matrix in the spanning family is $\mathcal{G}_F = (\text{Res}_F(b_r b_s))_{r,s=1}^\mu$. The $k=0$ term of (8.8) is the ordinary Gram matrix $\mathcal{G}_0$ of the residue pairing on B_0 , and all other terms have strictly positive support; classical local duality gives $\det \mathcal{G}_0 \neq 0$. Hence

$$\mathcal{G}_F = \mathcal{G}_0 + (\text{a positive-support matrix})$$

is invertible over K . If $\sum_r c_r b_r$ lies in (F) , pairing with each b_s gives $\mathcal{G}_F c = 0$, so all c_r vanish; thus the spanning family of [theorem 9.2](#) is a basis and (9.4) is perfect. Any decomposition modulo (F) has the same coordinates in a fixed basis, which gives independence of h . The quotients Q_j themselves need not be unique for $n > 1$. $\square$

The proof has a useful division of labour. A support-controlled Neumann operator prevents the dimension from increasing; a perturbed nondegenerate residue matrix prevents it from decreasing. Neither step assumes a simple zero, a triangular system, a polynomial perturbation, or a common radius of convergence for the perturbation coefficients. It is the last of these that is the point of this section: [theorem 9.3](#) applies to perturbations such as the coefficient family (3.9), which lies in $\mathcal{A}_1$ and in no $\mathcal{H}_\Gamma(U)$.

9.2 Multiplication matrices and residue reconstruction over $\mathcal{A}_n$

For $H \in \mathcal{A}_n$ define

$$M_H = (\text{NF}_F(Hb_s)_r)_{r,s=1}^\mu, \quad (9.5)$$

which is multiplication by the class of H in B_F ; hence $M_{H_1 H_2} = M_{H_1} M_{H_2}$ and $[M_{z_i}, M_{z_j}] = 0$. For ordinary basis elements and coordinate functions the entries have support in E^* and reduce at exponent zero to their ordinary counterparts. This gives explicit finite-dimensional data for a deformation whose coefficient support may be infinite and of arbitrary valuation rank.

The normal form is also reconstructible from residues. With $r_F(H) = (\text{Res}_F(Hb_s))_{s=1}^\mu$ and the column convention above,

$$\text{NF}_F(H) = \mathcal{G}_F^{-1} r_F(H), \quad (9.6)$$

the symmetry of $\mathcal{G}_F$ removing any transpose ambiguity. Since $\mathcal{G}_0$ is invertible, geometric matrix inversion shows $\mathcal{G}_F^{-1}$ has support in E^* , so (9.6) is a second construction with exactly the same support bound.

9.3 The common-domain form, over $\mathcal{R}_n$: a Koszul conjugacy

Over $\mathcal{R}_n$ a fixed ordinary domain is available, and with it a genuinely different proof which yields something the $\mathcal{A}_n$ argument does not: the vanishing of *all* higher homology of the Hahn Koszul complex. We record it.

Let $(C_\bullet, d)$ be a complex of complex vector spaces concentrated in nonnegative degrees with homology only in degree zero, and let $V = H_0(C_\bullet)$ be finite-dimensional. Choose linear maps $i : V \rightarrow C_0$, $p : C_0 \rightarrow V$ extended by zero in other degrees, and $h : C_k \rightarrow C_{k+1}$, with

$$pi = 1, \quad di = 0, \quad pd = 0, \quad dh + hd = 1 - ip, \quad h^2 = 0, \quad hi = 0, \quad ph = 0. \quad (9.7)$$

Such data always exist: split each space into boundaries and a complement mapping isomorphically onto the boundaries one degree lower, in degree zero also splitting off the selected representatives $i(V)$, and define h to invert d on boundaries and to vanish on the chosen complements and on $i(V)$. The construction can enforce any specified representatives of a basis of V . These are vector-space splittings: no boundedness or continuity of them is asserted. Their outputs nevertheless belong to the same specified space of holomorphic functions, which is the whole point.

Theorem 9.4 (Positive Hahn conjugacy). *Extend the data (9.7) coefficientwise to Hahn families. Let $\delta = \sum_{e \in E} t^e \delta_e$ be a degree-(-1) operator with $E \subset \Gamma_{>0}$ well ordered, and suppose $(d + \delta)^2 = 0$. Put $D = d + \delta$ and*

$$U = 1 + \delta h, \quad p_F = pU^{-1}, \quad h_F = hU^{-1}. \quad (9.8)$$

Then

$$D = UdU^{-1}, \quad Ui = i, \quad (9.9)$$

and

$$p_F i = 1, \quad p_F D = 0, \quad Dh_F + h_F D = 1 - ip_F. \quad (9.10)$$

All these operators are well defined, and applied to input support S the operators U^{-1}, p_F, h_F have output support in $S + E^$.*

Proof. The inverse of U is supplied by [theorem 2.5](#). Since the complex is concentrated in nonnegative degrees and $i(V)$ lies in degree zero, $\delta i = 0$. From $D^2 = 0$ we get $d\delta + \delta d + \delta^2 = 0$, and

$$Ud - DU = \delta h d - \delta - d\delta h - \delta^2 h = \delta h d - \delta + \delta d h = \delta(hd + dh - 1) = -\delta ip = 0,$$

so $Ud = DU$; and $Ui = i$ because $hi = 0$, giving (9.9). Since $h^2 = 0$ we have $Uh = h$ and hence $UhU^{-1} = h_F$; conjugating $dh + hd = 1 - ip$ by U gives the last identity in (9.10), and the other

two follow from $U^{-1}i = i$ and $U^{-1}D = dU^{-1}$. Every identity is an equality of strongly summable coefficients, justified by [theorem 2.3](#); the support bound follows from the geometric inverse for U together with the fact that p, h preserve support. $\square$

The absence of a transferred differential on V is not an extra assumption: it follows from concentration in degree zero. In a complex with homology in several degrees, positive perturbations can produce a nontrivial differential on the homology space, and the theorem would have to be modified.

Lemma 9.5 (Ordinary contraction on one fixed neighbourhood). *Let $f_1, \dots, f_n$ be ordinary holomorphic germs with an isolated common zero at the origin and local algebra $A_0 = R_n/(f)$ of dimension μ , and let $b_1, \dots, b_\mu$ be polynomial representatives of a basis. Choose a sufficiently small Stein polydisk Ω on which all f_i are holomorphic with no common zero except the origin, and form the global-section Koszul complex*

$$C_k = \mathcal{O}(\Omega) \otimes_{\mathbb{C}} \bigwedge^k \mathbb{C}^n, \quad d_f(g e_{i_1} \wedge \dots \wedge e_{i_k}) = \sum_{j=1}^k (-1)^{j-1} f_{i_j} g e_{i_1} \wedge \dots \widehat{e_{i_j}} \dots \wedge e_{i_k}. \quad (9.11)$$

Then (9.11) has no positive-degree homology and its degree-zero homology is A_0 ; it therefore admits a contraction (9.7) with $V = \mathbb{C}^\mu$ and $i(c_1, \dots, c_\mu) = \sum_j c_j b_j$. Every component of its maps takes holomorphic functions on Ω to holomorphic functions on that same Ω .

Proof. At the origin the n germs generate an ideal of finite colength in a regular local ring of dimension n , so they are a system of parameters and a regular sequence and their Koszul complex is exact in positive degrees. At any other point of Ω some f_i is a unit in the stalk and the Koszul complex is contractible. Thus the sheaf Koszul complex resolves a coherent quotient sheaf supported at the origin with stalk A_0 . Kernels in this finite complex are coherent; Cartan's theorem B on the Stein domain Ω makes their higher cohomology vanish, and applying it to the successive short exact sequences shows that global sections preserve the required exactness, the global sections of the quotient sheaf being A_0 . These ordinary analytic and regular-local facts are classical [14, 12, 19]. The vector-space splitting described above now supplies the maps, with codomains the fixed spaces (9.11). $\square$

This use of Cartan B is substantive, and it is exactly the step that has no analogue over $\mathcal{A}_n$: choosing a separate germ-level division for each coefficient and then intersecting all its neighbourhoods would not prove the final assertion of the lemma.

Theorem 9.6 (Support-explicit conservation of multiplicity over $\mathcal{R}_n$). *Let $f, \mu, b_1, \dots, b_\mu$ be as above, let $F_i = f_i + E_i$ with all $E_i \in \mathcal{R}_n$ of strictly positive Hahn support (zero errors allowed), and put $E = \bigcup_i \text{supp } E_i \subset \Gamma_{>0}$. Then:*

1. *the classes of $b_1, \dots, b_\mu$ form a K -basis of $A_F = \mathcal{R}_n/(F_1, \dots, F_n)$, so $\dim_K A_F = \mu$;*
2. *for $G \in \mathcal{R}_n$ there are K -linear normal-form coordinates $\text{NF}_F(G)_j$ and division certificates $Q_i(G) \in \mathcal{R}_n$ with*

$$G = \sum_{j=1}^{\mu} b_j \text{NF}_F(G)_j + \sum_{i=1}^n F_i Q_i(G), \quad (9.12)$$

and on one fixed ordinary domain representing the inputs these outputs have support in $\text{supp } G + E^$;*

3. *all higher homology of the Koszul complex of $F_1, \dots, F_n$ over $\mathcal{R}_n$ vanishes;*
4. *the multiplication matrices M_{z_ℓ} in the prescribed basis have entries supported in E^* , and their coefficients at exponent zero are the ordinary multiplication matrices of z_ℓ on A_0 .*

The normal-form coordinates are intrinsic once the basis is fixed; the division certificates may depend on the ordinary contraction chosen in the proof.

Proof. Take one small Stein polydisk Ω representing all errors, with the ordinary zero set of f equal to $\{0\}$ there, choose the contraction of [theorem 9.5](#), and extend it coefficientwise to the Hahn Koszul complex over $\mathcal{H}_\Gamma(\Omega)$. The perturbation $\delta = d_F - d_f$ is the Koszul differential formed from $E_1, \dots, E_n$; it has support in E and each coefficient is a linear operator on the same holomorphic function spaces. Koszul differentials square to zero, so [theorem 9.4](#) applies. In degree zero set

$$\mathrm{NF}_F(G) = p(1 + \delta h)^{-1}G, \quad Q(G) = h(1 + \delta h)^{-1}G \in C_1((t^\Gamma)). \quad (9.13)$$

The contraction identity gives $G = i \mathrm{NF}_F(G) + d_F Q(G)$, which is [\(9.12\)](#), and $\mathrm{NF}_F \circ i = 1$, $\mathrm{NF}_F \circ d_F = 0$; hence NF_F identifies the quotient with K^μ with basis the classes of the b_j . The conjugacy of complexes gives the positive-degree exactness. Every operation is performed on the one domain Ω , and [theorem 9.4](#) gives the support bound.

For an additional germ G , choose a smaller common Stein polydisk if needed and repeat; restrictions of the quotient algebras send each b_j to the same b_j and are isomorphisms because these are bases on both domains, so the normal forms agree under restriction. Passing to the direct limit proves the assertions over $\mathcal{R}_n$ and their independence from the chosen contraction, and positive-degree exactness likewise passes to the limit. Finally the j th column of M_{z_ℓ} is $\mathrm{NF}_F(z_\ell b_j)$; its input is ordinary, so its support is $\{0\}$ and the output support lies in E^* , and at exponent zero the inverse in [\(9.13\)](#) is the identity, so the column reduces to $p(z_\ell b_j)$. $\square$

9.4 A coefficient recursion, not an iteration limit

Formula [\(9.13\)](#) has an explicit reading. In degree zero put $L = \delta h = \sum_{e \in E} t^e L_e$ and $W = (1 + L)^{-1}G$; then

$$W_\nu = G_\nu - \sum_{\substack{e \in E, \eta \in \mathrm{supp} G + E^* \\ e + \eta = \nu}} L_e W_\eta, \quad (9.14)$$

where every η on the right is smaller than ν , and for fixed ν only finitely many input coefficients and finitely many words in the perturbation coefficients are involved. Then $\mathrm{NF}_{F,\nu} = pW_\nu$ and $Q_\nu = hW_\nu$. The analogous recursion over $\mathcal{A}_n$ is obtained from [\(9.2\)](#) with T in place of L .

This is an effective recursion *relative to* available ordinary division maps and an effective presentation of the support. Arbitrary complex-linear splittings and arbitrary well-ordered sets need not come with computable presentations, and the theorem does not assert a finite-time enumeration algorithm for every set-theoretically possible Hahn input. The finite-dependence statement at a specified coefficient, however, is unconditional.

Corollary 9.7 (Infinitesimal coordinate eigenvalues). *Over either ring, the matrices $M_{z_1}, \dots, M_{z_n}$ commute, and for each ℓ*

$$\det(XI - M_{z_\ell}) \equiv X^\mu \pmod{\mathfrak{m}_K[X]}. \quad (9.15)$$

Every coordinate eigenvalue of the deformed finite algebra is therefore infinitesimal.

Proof. The matrices represent multiplication in a commutative algebra, so they commute. In A_0 every coordinate lies in the nilpotent maximal ideal and its multiplication matrix is nilpotent with characteristic polynomial X^μ ; the reduction statement follows from the exponent-zero assertions of [theorem 9.6](#) over $\mathcal{R}_n$ and of [section 9.2](#) over $\mathcal{A}_n$. A root of a monic polynomial whose lower coefficients are infinitesimal must be infinitesimal, by the argument of [theorem 4.6](#). $\square$

10 Residue duality, traces, and exact zero multiplicities

10.1 The contour-expansion functional over $\mathcal{R}_n$

Over $\mathcal{R}_n$ one fixed ordinary residue cycle is available, and expanding on it produces a functional whose construction is visibly independent of the splitting used in [theorem 9.6](#). For an ordinary isolated complete intersection f in a small neighbourhood of the origin choose a standard Grothendieck residue cycle

$$\mathcal{C}_f = \{z : |f_1(z)| = \epsilon_1, \dots, |f_n(z)| = \epsilon_n\}$$

in a suitable sufficiently small finite-map neighbourhood, the radii chosen so that this is a compact regular real n -cycle oriented by $d \arg f_1 \wedge \dots \wedge d \arg f_n > 0$; equivalent cycles give the same integral, and the normalization is (8.3). On a neighbourhood of $\mathcal{C}_f$ every f_i is nonzero, so

$$\frac{1}{F_i} = \frac{1}{f_i} \sum_{k \geq 0} \left(-\frac{E_i}{f_i} \right)^k \quad (10.1)$$

is a legitimate common-support Hahn expansion there. For a uniformly supported Hahn family of differential forms on a compact ordinary cycle, define its H -integral coefficientwise; this is a coefficient functional, and the cycle is not being treated as a fine-continuous parametrized object in $\mathbf{SC}^n$. Define

$$\Lambda_F(G) = \frac{1}{(2\pi i)^n} \int_{\mathcal{C}_f}^H \frac{G dz_1 \wedge \dots \wedge dz_n}{F_1 \dots F_n}, \quad (10.2)$$

the form being expanded by (10.1) before its ordinary coefficients are integrated.

Theorem 10.1 (Support-controlled residue duality over $\mathcal{R}_n$). *For the data of [theorem 9.6](#), (10.2) is well defined and independent of the chosen sufficiently small ordinary residue cycle. It descends to a K -linear functional $\Lambda_F : A_F \rightarrow K$ with*

$$\text{supp } \Lambda_F(G) \subseteq \text{supp } G + E^*, \quad (10.3)$$

and the pairing $(u, v) \mapsto \Lambda_F(uv)$ on $A_F \times A_F$ is perfect. In the prescribed basis its Gram matrix $\mathcal{G}_F$ has support in E^* and satisfies

$$\mathcal{G}_F = \mathcal{G}_0 + (\text{positive-support terms}), \quad \mathcal{G}_0 = (\text{res}_f(b_r b_s))_{r,s}, \quad \det \mathcal{G}_0 \neq 0.$$

All these assertions remain valid when roots collide and the algebra becomes nonreduced.

Proof. Expanding all denominators gives coefficients supported in $\text{supp } G + E^*$ with finitely many contributions at each exponent, by [theorem 2.3](#); on the cycle each contribution is an ordinary differential form, so the integral exists coefficientwise with the stated support. The individual terms are ordinary Grothendieck residue expressions with denominators $f_1^{k_1+1} \dots f_n^{k_n+1}$, and their ordinary cycle independence gives independence of (10.2), including passage to smaller common neighbourhoods.

For $G = F_j H$, cancel the factor F_j before expanding: every resulting coefficient is an ordinary form with no f_j in its denominator, and its integral vanishes, since after adjoining a factor f_j to numerator and denominator it is a residue for the complete intersection of the indicated powers with numerator in its defining ideal. Thus Λ_F annihilates $(F_1, \dots, F_n)$; scalar Hahn linearity follows from finite convolution at each exponent.

For the Gram matrix, each $b_r b_s$ is ordinary, and at exponent zero the errors in the denominator expansion disappear, leaving exactly the classical matrix $\mathcal{G}_0$, whose determinant is nonzero by ordinary residue nondegeneracy. Hence $\det \mathcal{G}_F$ has nonzero coefficient $\det \mathcal{G}_0$ at exponent zero and $\mathcal{G}_F$ is invertible over K ; since the classes of the b_r form a basis of A_F by [theorem 9.6](#), the pairing is perfect. The argument never uses nonvanishing of a Jacobian at any root, which is the collision assertion. $\square$

Proposition 10.2 (The two residue functionals agree where both are defined). *Let $E_j \in \mathcal{R}_n$, so that both Res_F of (8.8) and Λ_F of (10.2) are defined on $\mathcal{R}_n$. Then $\text{Res}_F = \Lambda_F$.*

Proof. Expand (10.2) by (10.1) and integrate coefficientwise. The term indexed by $k \in \mathbb{N}^n$ is $(-1)^{|k|}$ times the ordinary integral of $G E_1^{k_1} \cdots E_n^{k_n} dz / (f_1^{k_1+1} \cdots f_n^{k_n+1})$ over $\mathcal{C}_f$, that is $(-1)^{|k|} \text{res}_{f^{k+1}}(GE^k)$ by (8.3). Summing over k is exactly (8.8), and the rearrangement is legitimate because [theorem 2.3](#) makes the family strongly summable. $\square$

Accordingly we write Res_F for both from now on, while remembering that over $\mathcal{A}_n$ only the series definition exists and no ordinary cycle carries all the coefficients.

Corollary 10.3 (A support-controlled constant residue frame). *Let $C = \mathcal{G}_0^{-1}(\mathcal{G}_F - \mathcal{G}_0)$. The matrix series*

$$S = (I + C)^{-1/2} = \sum_{k \geq 0} \binom{-1/2}{k} C^k \quad (10.4)$$

is strongly summable with support in E^ and satisfies $S^\top \mathcal{G}_F S = \mathcal{G}_0$. Thus the residue bilinear space has a basis congruent to the prescribed ordinary basis in which its matrix is exactly the ordinary residue matrix, even on the collision locus. The statement holds over $\mathcal{R}_n$ and, with $\mathcal{G}_F$ as in [theorem 9.3](#), over $\mathcal{A}_n$.*

Proof. The matrix C has positive support, so [theorem 2.3](#) makes every coefficient of (10.4) a finite sum, and ordinary one-variable formal power-series identities in the single matrix C give $S^2(I+C) = I$. Symmetry of $\mathcal{G}_F$ and $\mathcal{G}_0$ gives $C^\top \mathcal{G}_0 = \mathcal{G}_0 C$, hence $S^\top \mathcal{G}_0 = \mathcal{G}_0 S$. Since $\mathcal{G}_F = \mathcal{G}_0(I+C)$ and S commutes with C , the isometry follows. $\square$

This identifies bilinear spaces, not algebras. It does not erase the nonreduced algebraic structure of a collision; in particular a perfect residue pairing should not be confused with the usually degenerate trace pairing of a nonreduced finite algebra.

10.2 Finite-parameter testing without a uniform radius

The passage from ordinary identities to a Hahn identity needs care over $\mathcal{A}_n$, since an infinite perturbation cannot simply be assigned infinitely many small complex parameters and placed on one common contour. The following lemma is the substitute, and it is the exact place where [theorem 2.7](#) is used.

Lemma 10.4 (Finite dependency at a prescribed exponent). *Fix the leading system f , a positive perturbation-support set E , and a well-ordered numerator support S . At any prescribed exponent γ , every coefficient constructed from (9.2), (8.8), finite products and coefficientwise differentiation depends on only finitely many perturbation coefficient germs and finitely many numerator coefficient germs, and the contributing words have a finite maximum length.*

Proof. Each construction has an expansion whose contributions are indexed by a source exponent in S and a finite word in E . Finite products only concatenate finitely many such words, and differentiation changes coefficient germs but not their exponents. [Theorem 2.3](#) gives finitely many words producing γ , including a finite maximum length. Retaining exactly the source and perturbation coefficients occurring in them determines the coefficient in question. No assertion is made that there are only finitely many exponents below γ . $\square$

Theorem 10.5 (Trace–Jacobian identity). *Let B_F denote $\mathcal{A}_n/(F)$ under the hypotheses of [theorem 9.3](#), or $\mathcal{R}_n/(F)$ under those of [theorem 9.6](#). In either case, for every H in the ambient ring,*

$$\text{Res}_F(HJ_F) = \text{Tr}_K(M_H), \quad J_F = \det\left(\frac{\partial F_i}{\partial z_j}\right). \quad (10.5)$$

In particular $\text{Res}_F(J_F) = \mu$.

Proof. Fix a Hahn exponent γ . By [theorem 10.4](#) the coefficients at γ on both sides involve only finitely many ordinary analytic coefficient germs. Temporarily replace the associated perturbation monomials by independent ordinary complex parameters $u_1, \dots, u_q$, treating the finitely many numerator coefficients separately by linearity. This produces a finite analytic deformation $g_u = f + \sum_\ell u_\ell q_\ell$ of f — a finite family, whose germs do have a common ordinary representative neighbourhood even when the original Hahn family does not.

For that finite deformation, the ordinary cluster algebra is finite free with the chosen basis after restricting the parameter neighbourhood, and (8.6) holds; the justification is spelled out in [section A](#). The Taylor series in the parameters of its cluster residue is the geometric denominator expansion (8.8), since on a sufficiently small ordinary residue cycle the expansion converges uniformly. The formal parameter expansion of its normal-coordinate map is (9.2), respectively (9.13): that Neumann expression gives a decomposition in the parameter-adic completion, where the basis coordinates are unique. This identifies the expansion even when the chosen linear splitting h was not norm-continuous.

Now substitute $u_\ell = t^{e_\ell}$ for the finitely many retained perturbation exponents and extract the coefficient of t^γ . If several parameter monomials carry the same Hahn exponent their contributions are added, and there are finitely many by [theorem 2.3](#); omitted coefficient germs cannot contribute, by their selection. This proves equality at γ , and γ was arbitrary. Taking $H = 1$ gives the last assertion. $\square$

The point of this proof is not to rebrand the ordinary trace identity as new. It is to verify that the identity survives the absence of an Archimedean valuation scale, and — over $\mathcal{A}_n$ — the absence of a common ordinary radius.

10.3 Actual points account for the whole length

Theorem 10.6 (Conservation of singular zeros). *Let $Z_F = \{a \in \mathfrak{m}_K^n : F_1(a) = \dots = F_n(a) = 0\}$, over $\mathcal{A}_n$ under the hypotheses of [theorem 9.3](#) or over $\mathcal{R}_n$ under those of [theorem 9.6](#). Then Z_F is finite and nonempty, and there is an Artin-algebra decomposition*

$$B_F \simeq \prod_{a \in Z_F} B_a, \quad e_a = \dim_K B_a \geq 1, \quad \sum_{a \in Z_F} e_a = \mu, \quad (10.6)$$

with e_a given by (6.4). The points of Z_F are exactly the joint characters of the commuting multiplication matrices $M_{z_1}, \dots, M_{z_n}$, that is, their simultaneous eigenvalue tuples. A point is simple, equivalently $e_a = 1$, exactly when $J_F(a) \neq 0$.

Proof. B_F is a nonzero finite-dimensional commutative K -algebra, hence Artinian, and decomposes into its finitely many local factors. By theorem 5.2 over $\mathcal{A}_n$, respectively theorem 6.3 over $\mathcal{R}_n$, its maximal ideals are exactly the common zero evaluations with residue field K ; this gives (10.6). The factor at a is the quotient of the local ring at $\mathfrak{m}_a$ by the ideal generated by the F_i ; it has finite length, so completion does not change it, and theorem 5.4, respectively theorem 6.6, identifies the completed quotient, giving (6.4).

In the local factor at a , multiplication by z_j is a_j times the identity plus a nilpotent operator, so simultaneous triangularization of the commuting multiplication operators puts precisely these tuples on the diagonal with the corresponding local lengths; conversely a common eigenvector in the regular representation determines a character, and the evaluation description identifies that character with a point of Z_F .

For the simple-point criterion, work in the formal local quotient: its cotangent space is K^n modulo the rows of the Jacobian matrix at a . If that matrix is invertible the maximal ideal has zero cotangent space, hence is zero by Nakayama's lemma and the local algebra is K ; conversely a length-one local algebra is K with zero cotangent space, forcing the Jacobian to have rank n . $\square$

The theorem protects total *length*. It neither requires nor implies that the number of distinct points stays constant: a point of length μ can split into μ simple points or into intermediate nonreduced clusters.

Proposition 10.7 (Four equivalent forms of the simple-root criterion). *Over either ring, the following are equivalent: every point of Z_F has local multiplicity one; $J_F(a) \neq 0$ for all $a \in Z_F$; J_F is a unit in B_F ; the multiplication matrix M_{J_F} has nonzero determinant. When they hold, Z_F has exactly μ distinct points.*

Proof. The first equivalence is the last paragraph of theorem 10.6. An element of an Artinian algebra is a unit exactly when it lies in no maximal ideal, which gives the second; multiplication by an element of a finite-dimensional algebra is invertible exactly when that element is a unit, which gives the third. $\square$

Corollary 10.8 (Weighted evaluation and simple residues). *For every H in the ambient ring,*

$$\text{Res}_F(H J_F) = \sum_{a \in Z_F} e_a H(a), \quad (10.7)$$

and if all points are simple then

$$\text{Res}_F(H) = \sum_{a \in Z_F} \frac{H(a)}{J_F(a)}. \quad (10.8)$$

Proof. On B_a , multiplication by H is the scalar $H(a)$ plus a nilpotent operator, so its trace is $e_a H(a)$; summing and applying theorem 10.5 gives (10.7). If every point is simple then $B_F \simeq K^{Z_F}$; letting ϵ_a be the corresponding idempotent, (10.5) applied to ϵ_a gives $\text{Res}_F(\epsilon_a) J_F(a) = 1$, and expanding the class of H as $\sum_a H(a) \epsilon_a$ gives (10.8). Alternatively, apply (10.5) to H/J_F , which is legitimate by theorem 10.7. $\square$

Individual summands in (10.8) may be infinite surcomplex numbers; their finite sum is nevertheless the strongly defined residue (8.8), and the support of the total can be far smaller than the supports of the individual local contributions. Section 13 makes this cancellation explicit twice.

Remark 10.9 (What is and is not localized). At a multiple point (10.8) is not a valid formula with a zero denominator. The always-valid statement is the functional on the finite local algebra: the Artinian product decomposition restricts Res_F to a perfect functional on each local factor, their sum is the total functional, and the trace–Jacobian identity still holds. This gives an algebraic local residue interpretation without inventing small fine-topological integration cycles around individual nonstandard points.

11 Workspace invariance and the analytic–formal comparison

11.1 No additional roots at unrepresented surreal scales

Let $\Gamma \subseteq \Delta \subset \mathbf{No}$ be nonzero divisible set-sized ordered subgroups, and use subscripts to distinguish $K_\Gamma \subseteq K_\Delta$, $\mathcal{A}_{n,\Gamma} \subseteq \mathcal{A}_{n,\Delta}$ and $\mathcal{R}_n(\Gamma) \subseteq \mathcal{R}_n(\Delta)$. Both source manuscripts prove a result of this shape, over their own rings and by different arguments; we state both and compare the proofs rather than duplicating one of them.

Theorem 11.1 (Workspace invariance over $\mathcal{A}_n$). *For a system F as in theorem 9.3 with coefficients in $\mathcal{A}_{n,\Gamma}$ there is a natural isomorphism*

$$\frac{\mathcal{A}_{n,\Delta}}{(F)} \simeq K_\Delta \otimes_{K_\Gamma} \frac{\mathcal{A}_{n,\Gamma}}{(F)}. \quad (11.1)$$

The infinitesimal zero set is already contained in K_Γ^n and its local multiplicities do not change over K_Δ .

Proof. Use the same ordinary basis $b_1, \dots, b_\mu$ and the same splitting p, h in both workspaces. The proof of theorem 9.3 applies in each and gives this basis on each side. The multiplication constants computed by (9.2) for the products $b_r b_s$ already lie in K_Γ and are unchanged in K_Δ , so the canonical map in (11.1) identifies the bases and their products.

A point over K_Δ defines a character of this finite algebra; every coordinate of the point satisfies the characteristic polynomial of the corresponding coordinate multiplication matrix over K_Γ , and since K_Γ is algebraically closed all roots of that polynomial lie in K_Γ . Hence all coordinates already lie in the smaller field. Equivalently, base change of each finite local factor retains its residue field and its dimension, the nilpotent maximal ideal remaining nilpotent, which is preservation of points and multiplicities. $\square$

Theorem 11.2 (Base change and full surcomplex exhaustiveness over $\mathcal{R}_n$). *For the data of theorem 9.6 in $\mathcal{R}_n(\Gamma)$, with $K' = K_\Delta$, the natural map*

$$A_F(\Gamma) \otimes_K K' \xrightarrow{\sim} A_F(\Delta) \quad (11.2)$$

is an isomorphism. In the prescribed polynomial basis the multiplication matrices on both sides are identical. The zero set in $\mathfrak{m}_{K'}^n$ is exactly the original zero set in $\mathfrak{m}_K^n$ with unchanged local multiplicities. Consequently, evaluating the same coherent system on the entire surcomplex monad introduces no additional roots, and the count (10.6) is valid there as well.

Proof. The construction (9.13) depends only on the ordinary contraction and on the existing perturbation coefficients; enlarging the exponent group changes no coefficient calculation. By theorem 9.6 the same $b_1, \dots, b_\mu$ form a basis on each side of (11.2), and the natural map sends one basis to the other, hence is an algebra isomorphism. The characteristic polynomial of each multiplication matrix has coefficients in K and all its roots already lie in K by algebraic closedness, so any coordinate of a new common zero in K' would be one of these roots; evaluation then agrees with the original evaluation and no new point appears. Base change of the Artinian local factors preserves their dimensions and hence their lengths. Finally, the finitely many coordinates of any proposed surcomplex point have set-sized supports; enlarging Γ by those supports and taking the divisible subgroup they generate gives a set-sized Δ to which the preceding argument applies, by theorem 2.1. $\square$

Corollary 11.3 (Full-surcomplex conservation). *Interpret the equations of theorem 11.1 at arbitrary surcomplex tuples infinitesimal relative to $\mathbb{C}$. Every such common zero belongs to the original field K_Γ^n . Consequently the exact count $\sum_a e_a = \mu$ already accounts for all full-surcomplex infinitesimal zeros.*

Proof. Any proposed tuple lies in a set-sized divisible workspace K_Δ containing K_Γ , by theorem 2.1, and evaluation in it is compatible with the smaller workspace by the strong-sum definitions. Apply theorem 11.1. $\square$

Remark 11.4 (The two proofs, compared). Theorem 11.1 runs the $\mathcal{A}_n$ deformation argument twice, once in each workspace, and identifies the two resulting bases together with their multiplication constants; the base-change isomorphism is then a consequence of having the same basis on both sides. Theorem 11.2 instead observes that the single construction (9.13) is literally unchanged by enlarging Γ , because it reads only the ordinary contraction and the existing perturbation coefficients, and therefore yields the *identical* matrices rather than matched ones. The second formulation is the sharper statement about matrices; the first is available over the strictly larger ring. Both then close the argument the same way, by algebraic closedness of the smaller field. Neither is a corollary of the other, and neither may be transferred between rings.

This rules out a specific possible defect of a set-sized construction: an isolated cluster cannot acquire extra roots merely because a new surreal scale is allowed. The conclusion uses both finite dimensionality and algebraic closedness, and it is not a statement that arbitrary infinite-dimensional equations have workspace-independent solution sets. One further caution: the supports of individual root coordinates need not lie in E^* . Fractional scales such as $t^{1/2}$ may appear in individual roots even when every matrix coefficient uses only integral powers; divisibility of Γ accommodates these scales, and section 13.2 exhibits them.

11.2 Allowing divergent ordinary coefficient series

There is also a useful comparison with the formal-coefficient ring $\mathcal{F}_n = \mathbb{C}[[z_1, \dots, z_n]]((t^\Gamma))$, whose coefficients are arbitrary formal complex power series. Strong evaluation still makes sense at every infinitesimal tuple, but coefficientwise ordinary analyticity has been dropped. The following theorem separates that change precisely from completion at a point. It is a statement about $\mathcal{A}_n$ and $\mathcal{F}_n$ only; no analogue for $\mathcal{R}_n$ is claimed.

Theorem 11.5 (Analytic–formal comparison). *The inclusion $\mathcal{A}_n \hookrightarrow \mathcal{F}_n$ is faithfully flat. Both rings have the same evaluation maximal ideals, the same residue fields, and the same completed local*

rings at corresponding points. If $I \subseteq \mathcal{A}_n$ has finite-dimensional quotient over K , the natural map

$$\mathcal{A}_n/I \longrightarrow \mathcal{F}_n/I\mathcal{F}_n \quad (11.3)$$

is an isomorphism. In particular, allowing divergent ordinary coefficient germs does not change a finite zero scheme defined over $\mathcal{A}_n$.

Proof. The arguments of sections 3 to 5 work with $\mathbb{C}[[z]]$ in place of $\mathbb{C}\{z\}$, using formal Weierstrass division; translation, finite jets and faithful evaluation use only formal Taylor identities and the support lemma. Consequently $\mathcal{F}_n$ is Noetherian, its maximal ideals are the same coordinate evaluation ideals, and its local completion at a is $K[[X]]$.

Let $A = (\mathcal{A}_n)_{\mathfrak{m}_a}$ and $B = (\mathcal{F}_n)_{\mathfrak{m}_a}$. The induced map of completions is the identity on $K[[X]]$, and completion of a Noetherian local ring is faithfully flat [19, tag 00MC]. It follows that $A \rightarrow B$ is flat: tensoring a flatness obstruction $\mathrm{Tor}_1^A(M, B)$ with the faithfully flat B -module $\widehat{B}$ turns it into $\mathrm{Tor}_1^A(M, \widehat{B}) = \mathrm{Tor}_1^A(M, \widehat{A}) = 0$, and faithfulness of $B \rightarrow \widehat{B}$ forces the obstruction to vanish. Flatness may be checked at maximal ideals of the target, so $\mathcal{A}_n \rightarrow \mathcal{F}_n$ is flat, and every maximal ideal of the source has its corresponding evaluation ideal in the target, making the map faithfully flat.

Now assume $\dim_K \mathcal{A}_n/I < \infty$. Its Artin decomposition has finitely many supporting points $a_1, \dots, a_r$, and for some N the product $J = \prod_{\ell=1}^r \mathfrak{m}_{a_\ell}^N$ is contained in I . The ideals for distinct points are comaximal, so by the Chinese remainder theorem and the identical finite-jet quotients of the two rings the map $\mathcal{A}_n/J \rightarrow \mathcal{F}_n/J\mathcal{F}_n$ is an isomorphism; quotienting by the image of I/J gives (11.3). $\square$

By theorem 3.11 the inclusion is nevertheless strict, so the comparison has content beyond a change of notation. Together with theorem 3.10 it completes the picture promised in section 1.2: the passage from common-domain to radius-free coefficients is a genuine enlargement that no common-domain theorem covers, whereas the further passage to formal coefficients, although also a genuine enlargement of the ring, changes nothing about finite-colength quotients. The comparison also provides a precise bridge to formal or full separated Hahn analytic structures without asserting that the two coefficient rings are equal.

12 Quantitative Hensel–Rouché stability over $\mathcal{A}_n$

Conservation of total length does not by itself label roots after a perturbation: a singular point may split, and several simple points may collide. This section obtains a root-by-root statement over $\mathcal{A}_n$ with an explicit threshold, formulated through the valuation loss in the inverse Jacobian, which can be substantially better than a bound using only its determinant.

12.1 A local estimate at a simple root

Assume $F \in \mathcal{A}_n^n$ with $v_H(F_i) \geq 0$ and $F(a) = 0$ for $a \in \mathfrak{m}_K^n$. If $L = DF(a)$ is invertible, define

$$\kappa_a = \max\left(0, -\min_{i,j} v((L^{-1})_{ij})\right), \quad (12.1)$$

so every entry of L^{-1} has valuation at least $-\kappa_a$, while every entry of L has nonnegative valuation. The adjugate formula yields the convenient but *sometimes strictly weaker* bound

$$0 \leq \kappa_a \leq v(\det L). \quad (12.2)$$

The inequality lives in Γ , an arbitrary ordered group; it is not an estimate obtained by replacing valuations with real-valued magnitudes.

Lemma 12.1 (Scaled Taylor polynomials at each Hahn exponent). *For $a \in \mathfrak{m}_K^n$, $s \in \Gamma_{>0}$ and $F \in \mathcal{A}_n$, the expansion $F(a + t^s y)$ has polynomial complex coefficient functions in y at every fixed Hahn exponent. Consequently it may be translated by any fixed complex vector in the y variables.*

Proof. First translate by a using [theorem 3.3](#), then expand in monomials in $t^s y$. At a fixed exponent γ the possible Taylor degrees and source exponents satisfy $\lambda + rs = \gamma$, and [theorem 2.3](#) gives finitely many such decompositions, hence finitely many degrees; each degree has finitely many multivariate monomials. The resulting coefficient is therefore a polynomial, so a fixed complex translation is legitimate coefficientwise and preserves Hahn support. $\square$

Theorem 12.2 (One-root Hensel–Rouché bound over $\mathcal{A}_n$). *Under the hypotheses above let $G = F + D$ with $D \in \mathcal{A}_n^n$ nonzero and $\delta = \min_i v_H(D_i) > 2\kappa_a$. Then G has a unique zero b in the ball*

$$\mathcal{B}_a = \{b \in \mathfrak{m}_K^n : v_{\min}(b - a) > \kappa_a\}, \quad (12.3)$$

it satisfies

$$v_{\min}(b - a) \geq \delta - \kappa_a, \quad (12.4)$$

and it is simple.

Proof. Put $\kappa = \kappa_a$ and $s = \delta - \kappa > \kappa$, and introduce the normalized system $H(y) = t^{-s} L^{-1} G(a + t^s y)$. Its linear contribution from F is exactly y . Since all Taylor coefficients of F at a have valuation at least zero, the terms of degree at least two in the normalized contribution have coefficients of valuation at least $2s - s - \kappa = s - \kappa = \delta - 2\kappa > 0$. The constant vector $c = t^{-s} L^{-1} D(a)$ has valuation at least zero, and every nonconstant contribution of D has coefficient valuation at least $\delta - \kappa = s > 0$. Thus

$$H(y) = y + c_0 + P(y), \quad c_0 = \text{red}(c) \in \mathbb{C}^n, \quad v_H(P_i) > 0. \quad (12.5)$$

By [theorem 12.1](#) the coefficient functions here are polynomials, so the substitution $y = -c_0 + u$ is valid even though the original coefficients were only germs; the resulting system is $u + \tilde{P}(u)$ with strictly positive perturbation support. Applying [theorems 9.3](#) and [10.6](#) to the ordinary leading system $(u_1, \dots, u_n)$, of length one, gives exactly one infinitesimal zero u . Hence $b = a + t^s(-c_0 + u)$ is a zero of G , lies in $\mathfrak{m}_K^n$ and satisfies (12.4); the length-one assertion makes it simple, since the affine change of variables and the invertible normalizing matrix preserve nonvanishing of the Jacobian.

It remains to prove uniqueness in the whole ball (12.3), not merely in the residue class used to construct the zero. Suppose $a + h$ and $a + k$ are zeros with $v_{\min}(h), v_{\min}(k) > \kappa$. Formal Taylor subtraction, factoring each difference of monomials, gives $G(a + h) - G(a + k) = (L + C)(h - k)$ where every entry of C has valuation at least $\min\{v_{\min}(h), v_{\min}(k), \delta\} > \kappa$: terms from F beyond its linear term retain at least one factor from h or k after subtraction, and terms from D retain valuation at least δ . All sums are strongly defined by [theorem 2.3](#). Thus $L^{-1}C$ has entries of strictly positive valuation, so $I + L^{-1}C$ is invertible by strong matrix geometric inversion, and the difference equation forces $h = k$. $\square$

The construction proves existence inside K itself. It does not invoke sequential completeness or a rank-one Banach contraction theorem, and all normalizations use actual monomials t^s with $s \in \Gamma$.

12.2 Correspondence for an entire finite cluster

Corollary 12.3 (Separation of simple zeros). *If $a \neq a'$ are two simple infinitesimal zeros of the same nonnegative-support system $F \in \mathcal{A}_n^n$, then*

$$v_{\min}(a - a') \leq \min(\kappa_a, \kappa_{a'}), \quad (12.6)$$

and the balls $\mathcal{B}_a$ around the simple zeros are pairwise disjoint.

Proof. The uniqueness argument of [theorem 12.2](#) with $D = 0$ says a is the only zero of F with $v_{\min}(b - a) > \kappa_a$; applying it at both points gives (12.6). If a point lay in both balls the ultrametric inequality would give $v_{\min}(a - a') > \min(\kappa_a, \kappa_{a'})$, a contradiction. $\square$

Theorem 12.4 (Finite-cluster correspondence). *Let F satisfy [theorem 9.3](#) with all μ zeros simple, and let $G = F + D$ with*

$$\delta = \min_i v_H(D_i) > 2 \max_{a \in Z_F} \kappa_a. \quad (12.7)$$

Then G also has μ simple infinitesimal zeros, and there is a unique bijection $a \mapsto b_a$ from Z_F to Z_G with $v_{\min}(b_a - a) > \kappa_a$ and $v_{\min}(b_a - a) \geq \delta - \kappa_a$. The same assertion accounts for all infinitesimal surcomplex zeros.

Proof. The case $D = 0$ is immediate. Otherwise each ball contains a unique simple zero of G by [theorem 12.2](#), and the balls are disjoint by [theorem 12.3](#), so μ distinct simple zeros have been found. Since G is still a positive-support perturbation of the same leading system f , [theorem 10.6](#) gives total length μ , so there are no others. The full-surcomplex assertion follows from [theorem 11.3](#), applied in a workspace containing both systems. $\square$

Example 12.5 (Sharpness of the strict threshold). Fix $\eta > 0$ and set $F(z) = z^2 - t^{2\eta}$, $D(z) = t^{2\eta}$, $G(z) = z^2$. The two roots of F are $a = \pm t^\eta$ and $\kappa_a = \eta$. Here $\delta = 2\eta = 2\kappa_a$, yet the two simple roots coalesce into a double root, and the displacement to zero has valuation exactly η , not greater than κ_a . Thus replacing the strict inequality in (12.7) by equality is false, and any universal factor smaller than two is false as well. This is the familiar sharp scalar Hensel obstruction, situated here in the analytic setting.

For higher-rank groups the threshold is an inequality in Γ , not an estimate after replacing valuation by a real-valued magnitude. With $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$, a perturbation of valuation N for arbitrarily large natural N still need not be smaller than a separation scale governed by ω .

Remark 12.6 (Two certificates of stability, and which is sharper for which purpose). The material of this section straddles two reports and is stated in full in the companion report on finite deformations [10], which carries the complementary certificate as well. We record the comparison here because [theorem 12.2](#) is proved over $\mathcal{A}_n$ and belongs to this report, and because averaging the two available certificates into a single “discriminant or Jacobian threshold” would lose real content.

1. The inverse-Jacobian certificate above is the one to use when individual roots are to be *matched*: it supplies the bijection $a \mapsto b_a$ of [theorem 12.4](#) together with the displacement bound (12.4). A companion manuscript proves in addition that both valuation bounds are attained, including the second-order bound $v(h + A^{-1}\Delta(a)) \geq 2\sigma - 3\kappa$; that attainment statement is proved there and is not re-proved here.

2. A second certificate in that cluster *declines root matching altogether*, on the explicit ground that a labelling of individual roots need not be Lipschitz, that roots of a multiple leading system generally involve fractional exponents, and that inverse Jacobians can have negative valuation. It certifies stability instead through a Discriminant–Jacobian identity and a discriminant threshold, offering “one certified geometric consequence without choosing root branches.” That certificate is the right one when only the statement that all zeros remain simple is needed and no root labelling is available.
3. Within the present section, (12.2) shows the determinant-only bound is provably lossy: [section 13.1](#) exhibits a system where $\kappa_a = \max(\alpha, \beta)$ while $v(\det L) = \alpha + \beta$, so the threshold $\delta > 2\max(\alpha, \beta)$ is strictly better than $\delta > 2(\alpha + \beta)$.

These are complementary certificates of different things, not competing estimates of one thing.

13 Worked examples

The first two examples are systems over $\mathcal{A}_2$; the second of them lies outside $\mathcal{R}_2$ and is the point of the radius-free ring. The third and fourth are systems over $\mathcal{R}_2$ and exhibit a collision and a nonpolynomial, non-grid support.

13.1 Four coupled roots and a perfect residue matrix, over $\mathcal{A}_2$

Let $A, B \in \mathfrak{m}_K$ be nonzero and consider

$$F_1 = x^2 - Ay, \quad F_2 = y^2 - Bx. \quad (13.1)$$

The ordinary leading ideal is (x^2, y^2) with basis $1, x, y, xy$ and length four, and [theorem 9.3](#) gives the same basis for the perturbed quotient without eliminating variables. Put $q = AB$; in the quotient

$$x^2 = Ay, \quad y^2 = Bx, \quad x^2y = qx, \quad xy^2 = qy, \quad (xy)^2 = qxy.$$

With columns indexed by multiplication of the ordered basis $(1, x, y, xy)$,

$$M_x = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & q \\ 0 & A & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad M_y = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & B & 0 \\ 1 & 0 & 0 & q \\ 0 & 1 & 0 & 0 \end{pmatrix}. \quad (13.2)$$

They commute and satisfy the two defining relations, and

$$\chi_x(\lambda) = \lambda(\lambda^3 - A^2B), \quad \chi_y(\lambda) = \lambda(\lambda^3 - AB^2). \quad (13.3)$$

The Hahn residue expansion is explicit. From (8.8) and (8.4),

$$\text{Res}_F(H) = \sum_{k, \ell \geq 0} A^k B^\ell [x^{2k+1-\ell} y^{2\ell+1-k}] H, \quad (13.4)$$

a coefficient with a negative index being zero; the two signs from the negative perturbations cancel the factor $(-1)^{k+\ell}$. For a monomial $x^p y^r$ the only possible indices are $k = (2p + r - 3)/3$ and

$\ell = (p + 2r - 3)/3$, so its residue is $A^k B^\ell$ when these are nonnegative integers and zero otherwise. In particular $\text{Res}_F(1) = \text{Res}_F(x) = \text{Res}_F(y) = 0$ and $\text{Res}_F(xy) = 1$, whence

$$\mathcal{G}_F = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & q \end{pmatrix}, \quad \det \mathcal{G}_F = 1. \quad (13.5)$$

Its determinant remains a unit even as the points collide under specialization of the parameters. This is why the residue pairing, rather than a list of distinct roots, is the stable invariant.

Choose $r \in K$ with $r^3 = A^2 B$ and let $\zeta \in \mathbb{C}$ be a primitive cube root of unity. The four roots are

$$(0, 0), \quad \left(r\zeta^j, \frac{r^2}{A}\zeta^{2j}\right), \quad j = 0, 1, 2, \quad (13.6)$$

since a nonzero solution has $y = x^2/A$ and hence $x^3 = A^2 B$, and conversely. Writing $\alpha = v(A) > 0$ and $\beta = v(B) > 0$, the nonzero coordinate valuations are

$$v(x) = \frac{2\alpha + \beta}{3}, \quad v(y) = \frac{\alpha + 2\beta}{3}, \quad (13.7)$$

all infinitesimal, with no assumption that α and β are Archimedean-equivalent. The Jacobian is $J_F = 4xy - q$, equal to $-q$ at the origin and to $3q$ at every nonzero root, so all four points are simple. Their residue weights for $H = 1$ are $-1/q$, $1/(3q)$, $1/(3q)$, $1/(3q)$ and sum to zero, each being infinite relative to $\mathbb{C}$; for $H = xy$ the sum is 1, and $\text{Res}_F(J_F) = 4$ while $\text{Tr}(M_{xy}) = 3q$, in agreement with (13.4) and theorem 10.5. Taking $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$, $A = t$ and $B = t^\omega$, the nonzero coordinates have valuations $(2 + \omega)/3$ and $(1 + 2\omega)/3$ and every nonzero residue weight has valuation $-(1 + \omega)$; replacing this value group by a single real-valued small parameter would discard the fact that ω dominates every positive integer.

The inverse-Jacobian loss is computable exactly here. At zero

$$DF(0)^{-1} = \begin{pmatrix} 0 & -B^{-1} \\ -A^{-1} & 0 \end{pmatrix}, \quad \text{and at a nonzero root} \quad DF(a)^{-1} = \frac{1}{3AB} \begin{pmatrix} 2y & A \\ B & 2x \end{pmatrix}.$$

With (13.7) this gives, at all four roots,

$$\kappa_a = \max(\alpha, \beta), \quad v(J_F(a)) = \alpha + \beta. \quad (13.8)$$

Thus a further perturbation with $\delta > 2 \max(\alpha, \beta)$ preserves all four roots individually with the displacement bound of theorem 12.4, and the inverse-matrix threshold is strictly better than the determinant-only sufficient condition $\delta > 2(\alpha + \beta)$, as promised in theorem 12.6.

13.2 A genuinely radius-free, higher-rank deformation over $\mathcal{A}_2$

Take $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$ and consider the analytic-germ system

$$F_1 = x^2 + \sum_{r \geq 1} \frac{t^r}{1 - rx - r^2 y}, \quad F_2 = y^2 + t^\omega x. \quad (13.9)$$

The family of coefficient germs in F_1 has *no* common ordinary neighbourhood of the origin: restricted to $y = 0$ it has poles at $x = 1/r$. Hence (13.9) is a system over $\mathcal{A}_2$ that lies in no $\mathcal{H}_\Gamma(U)$, and

[theorem 9.6](#) does not apply to it; this is precisely the gap that [theorem 9.3](#) fills. Its support is the well-ordered set of positive integers and the combined perturbation support lies in $\mathbb{N}_{>0} \cup \{\omega\}$, so [theorem 9.3](#) applies and gives total length four on the full surcomplex infinitesimal neighbourhood, by [theorem 11.3](#).

More can be read off without solving the infinite equations. Make the diagonal rescaling

$$x = t^{1/2}u, \quad y = t^{\omega/2+1/4}v, \quad (13.10)$$

divide the first equation by t and the second by $t^{\omega+1/2}$. The resulting leading system is

$$u^2 + 1 = 0, \quad v^2 + u = 0, \quad (13.11)$$

and every remaining coefficient has strictly positive Hahn valuation. After this rescaling each Hahn coefficient is a polynomial in (u, v) , by the coordinatewise version of [theorem 12.1](#), so we may translate to each of the four ordinary solutions (u_0, v_0) of (13.11). At each one

$$u_0 = \pm i, \quad v_0^2 = -u_0, \quad \det D(u^2 + 1, v^2 + u) = 4u_0v_0 \neq 0,$$

so the length-one case of [theorem 9.3](#) supplies one infinitesimal lift in each translated chart; the charts are disjoint by their different complex reductions, and since the unscaled system has total length four these are all the zeros and all are simple. Their leading coordinates are

$$x = u_0 t^{1/2} + \dots, \quad y = v_0 t^{\omega/2+1/4} + \dots, \quad (13.12)$$

with larger-valuation corrections, and at such a root

$$J_F = 4u_0v_0 t^{\omega/2+3/4} + \dots : \quad (13.13)$$

the derivative of the positive perturbation in F_1 has valuation at least 1, whereas $v(x) = 1/2$ and $v(y) = \omega/2 + 1/4$, so $4xy$ dominates the remaining terms of the determinant. Note that the root supports involve $t^{1/2}$ and $t^{\omega/2+1/4}$, which lie in no monoid generated by the perturbation support; this is the caution recorded after [theorem 11.4](#).

Although every individual simple residue $1/J_F(a)$ is infinite, their sum is not merely finite — it is infinitesimal:

$$\boxed{\text{Res}_F(1) = -4t + R, \quad v(R) > 1.} \quad (13.14)$$

For this computation use $f = (x^2, y^2)$ in (8.8). The exponent-zero contribution vanishes. A contribution at exponent 1 can come only from $k = 1, \ell = 0$ and the coefficient $1/(1 - x - y)$ of t in the first perturbation, and it equals

$$-[x^3y] \frac{1}{1 - x - y} = -4.$$

Every other nonzero contribution has valuation strictly greater than 1; terms involving the second perturbation have valuation at least ω . Equation (10.8) identifies (13.14) with the sum over the four infinite local weights. This single example simultaneously uses shrinking ordinary radii, several-variable coupling, different valuation ranks, and cancellation beyond the leading pole scale.

13.3 A length-five collision with three scale regimes, over $\mathcal{R}_2$

Let $s, u \in \mathfrak{m}_K$ and consider

$$F_1 = x^2 - y^3 - s, \quad F_2 = xy - u. \quad (13.15)$$

At $s = u = 0$ the ordinary zero is isolated, and its local algebra has basis

$$(b_1, \dots, b_5) = (1, y, y^2, y^3, x), \quad \mu = 5, \quad (13.16)$$

since the leading monomials of a local polynomial reduction are x^2, xy, y^4 . The defining relations imply

$$xy = u, \quad x^2 = s + y^3, \quad y^4 = ux - sy, \quad (13.17)$$

so with columns representing images of the ordered basis (13.16),

$$M_x = \begin{pmatrix} 0 & u & 0 & 0 & s \\ 0 & 0 & u & 0 & 0 \\ 0 & 0 & 0 & u & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad M_y = \begin{pmatrix} 0 & 0 & 0 & 0 & u \\ 1 & 0 & 0 & -s & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & u & 0 \end{pmatrix}, \quad (13.18)$$

satisfying exactly

$$M_x M_y = M_y M_x = uI, \quad M_x^2 - M_y^3 = sI. \quad (13.19)$$

Starting from the vector for 1 these matrices generate the five basis vectors, so they give a faithful five-dimensional realization of the algebra described by (13.17); [theorem 9.6](#) supplies the same conclusion for the analytic monad quotient. Their characteristic polynomials are

$$\chi_y(Y) = Y^5 + sY^2 - u^2, \quad \chi_x(X) = X^5 - sX^3 - u^3. \quad (13.20)$$

When $u \neq 0$ every root of χ_y is nonzero and determines one solution by $x = u/y$. In all cases the total local length is five: the polynomial equations are elimination tools, and a chosen coordinate need not distinguish all points.

The actual collision equation. The Jacobian is

$$J_F = \det \begin{pmatrix} 2x & -3y^2 \\ y & x \end{pmatrix} = 2x^2 + 3y^3 = 2s + 5y^3 \quad \text{in } A_F, \quad (13.21)$$

and a direct calculation gives

$$\det(M_{J_F}) = 3125u^6 - 108s^5. \quad (13.22)$$

By [theorem 10.7](#) the system therefore has five simple points exactly when the right-hand side is nonzero. This is the collision equation of the *whole algebra*, not the discriminant of a projected coordinate: indeed

$$\text{disc}_Y(Y^5 + sY^2 - u^2) = u^2(3125u^6 - 108s^5), \quad (13.23)$$

and the extra factor u^2 is a projection artifact. At $u = 0, s \neq 0$ the system has the two points $(\pm\sqrt{s}, 0)$ and the three points $(0, y)$ with $y^3 = -s$, all five simple, even though the polynomial in y has a double root at zero. This is why simultaneous multiplication matrices and the full Jacobian give better finite-map data than a single eliminant.

A nontrivial collision is explicit. Choose $a \in \mathfrak{m}_K^\times$, choose $\eta \in \mathbb{C}$ with $\eta^2 = -3/2$, and put

$$s = -\frac{5}{2}a^6, \quad u = \eta a^5.$$

Then $(x, y) = (\eta a^3, a^2)$ is a zero with $J_F = 0$, and (13.22) vanishes. In the y -eliminant this is a double root, its second derivative there being $15a^6 \neq 0$: writing $y = a^2Y$, the eliminant is proportional to $(Y - 1)^2(2Y^3 + 4Y^2 + 6Y + 3)$, whose greatest common divisor with its derivative is $Y - 1$. This specialization therefore has one point of length two and three other simple points, while at $s = u = 0$ all five units of length are concentrated at the origin.

Three infinitesimal scale regimes. Take positive $\alpha, \beta \in \Gamma$ and set $s = t^\alpha$, $u = t^\beta$; the relative sizes are governed by 5α and 6β . All roots are simple, since in the equal case the two leading coefficients in (13.22) subtract to $3125 - 108 = 3017 \neq 0$.

Regime	Number	$v(y)$	$v(x)$
$5\alpha < 6\beta$	2	$\beta - \alpha/2$	$\alpha/2$
	3	$\alpha/3$	$\beta - \alpha/3$
$5\alpha > 6\beta$	5	$2\beta/5$	$3\beta/5$
$5\alpha = 6\beta$	5	$2\beta/5$	$3\beta/5$

These assertions are proved directly by rescaling rather than by displaying a Newton polygon. Suppose $5\alpha < 6\beta$. Substituting $y = t^{\beta-\alpha/2}Y$ in $\chi_y = 0$ and dividing by $t^{2\beta}$ gives

$$Y^2 - 1 + t^{3\beta-5\alpha/2}Y^5 = 0,$$

whose error exponent is positive; each of the two simple ordinary roots $Y = \pm 1$ lifts uniquely to a root in its monad, by degree-one preparation or by the simple-root case of theorem 10.6. Next put $y = t^{\alpha/3}Y$ and divide by $t^{5\alpha/3}$, obtaining

$$Y^5 + Y^2 - t^{2\beta-5\alpha/3} = 0,$$

whose three nonzero simple leading roots satisfy $Y^3 = -1$ and lift to the second group. The valuations in the two groups differ and together they account for all five roots; since $x = t^\beta/y$, the stated x -valuations follow. For $5\alpha > 6\beta$, put $y = t^{2\beta/5}Y$ and divide by $t^{2\beta}$ to get $Y^5 - 1 + t^{\alpha-6\beta/5}Y^2 = 0$, whose five ordinary fifth roots of unity are simple and lift uniquely. In the equal case the same substitution gives the ordinary polynomial $Y^5 + Y^2 - 1$, which has no zero root and is squarefree, its discriminant being 3017; its five ordinary roots give the five solutions at that scale. This proves the table in every divisible ordered value group, not merely for real rational exponents.

Residues without singular denominators. In the basis (13.16) the total residue functional is

$$\text{Res}_F(1) = \text{Res}_F(y) = \text{Res}_F(y^2) = \text{Res}_F(x) = 0, \quad \text{Res}_F(y^3) = 1. \quad (13.24)$$

One way to identify it first on the simple locus with $u \neq 0$ is to use the y -eliminant $q(Y) = Y^5 + sY^2 - u^2$, for which $q'(y) = yJ_F$ at a root; the classical coefficient identity for a monic quintic gives

$$\sum_{q(y)=0} \frac{y^k}{q'(y)} = 0 \quad (0 \leq k < 4), \quad \sum_{q(y)=0} \frac{y^4}{q'(y)} = 1,$$

and these together with $x = u/y$ give (13.24). The fixed-cycle residue depends holomorphically on the ordinary parameters near $(0, 0)$, so equality on the open simple locus proves the identities throughout the parameter germ and hence at every infinitesimal Hahn specialization; they may also be obtained directly from the ordinary residue expansion (10.1). The Gram matrix is

$$\mathcal{G}_F = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -s & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad \det \mathcal{G}_F = 1. \quad (13.25)$$

Unlike (13.22) this determinant never vanishes: the residue pairing remains perfect at every collision, including the length-five point at the origin. By contrast the trace Gram matrix is $\mathcal{G}_F M_{J_F}$, so

$$\det(\mathrm{Tr}(M_{b_r b_s}))_{r,s} = 3125u^6 - 108s^5$$

does degenerate on the collision locus — the distinction recorded after theorem 10.3. Finally $\mathrm{Res}_F(J_F) = \mathrm{Res}_F(2s + 5y^3) = 5$, the exact count, and on the simple locus

$$\sum_{F(a)=0} \frac{1}{J_F(a)} = 0, \quad \sum_{F(a)=0} \frac{y(a)^3}{J_F(a)} = 1, \quad \sum_{F(a)=0} 1 = 5. \quad (13.26)$$

For monomial parameters in the scale table each $J_F(a)$ is a nonzero infinitesimal, so every summand of the first sum has infinite modulus; their cancellation is nevertheless exact, and the total residue is computed by a fixed nonsingular matrix rather than by subtracting poorly separated pole contributions.

13.4 Beyond polynomial and grid-supported perturbations, over $\mathcal{R}_2$

The finite example is not the limit of the theorem's scope. Consider the common-domain entire coefficient families

$$\begin{aligned} F_1(x, y) &= x^2 - y^3 - \sum_{m \geq 1} t^{2-1/m} \exp(m! x), \\ F_2(x, y) &= xy - t^{\sqrt{2}} - t^3 \sin(x + y). \end{aligned} \quad (13.27)$$

Choose a divisible workspace containing the real exponents involved. The support

$$E = \{2 - 1/m : m \geq 1\} \cup \{\sqrt{2}, 3\}$$

is positive and well ordered; its subset $\{2 - 1/m\}$ has a finite upper cut at 2 while its denominators are unbounded, so the construction is not restricted to a discrete one-generator grid — this is the situation described abstractly in theorem 2.7. All coefficient functions are entire, so a common ordinary domain is immediate despite their very large growth.

Corollary 13.1 (A nonpolynomial length-five cluster over $\mathcal{R}_2$). *The system (13.27) has finitely many common zeros in the full surcomplex monad of $(0, 0)$, of total local multiplicity five. Its quotient algebra has the exact basis $(1, y, y^2, y^3, x)$. Every entry of its two multiplication matrices and of its residue Gram matrix has support contained in E^* , the residue Gram matrix is invertible, and the total residue of its Jacobian equals five.*

Proof. Its ordinary leading system is exactly $(x^2 - y^3, xy)$, of multiplicity five. Apply [theorems 9.6, 10.1, 10.5, 10.6](#) and [11.2](#). Each input to a multiplication or Gram matrix entry is ordinary, giving the support bound E^* . $\square$

This corollary does not assert that the explicit polynomial matrices [\(13.18\)](#) remain valid for the nonpolynomial system; they do not. It asserts that the same *basis size and basis representatives* survive and that the general normal-form recursion [\(9.14\)](#) computes the new entries. Nor does it assert that all five roots are simple without computing the new Jacobian determinant in that algebra. The proof is unaffected by replacing E with a positive well-ordered set in a higher-rank subgroup of **No**, or by using several distinct Archimedean classes of scales: the only summation requirements are well-ordering and finite contribution at each exponent, and no ordinary bound on coefficient derivatives is required.

14 Hypothesis audit, limits, and open questions

14.1 Hypotheses that cannot be dropped silently

An isolated complete intersection. The leading system must have finite local colength, and exactly n equations in n variables are required. A leading system such as (xy, xy) in two variables has a positive-dimensional zero set, so no finite μ is available; the quotient by the single equation x in two variables has infinite dimension over the coefficient field, and there is no finite length to conserve. Arbitrary overdetermined perturbations can *lower* length: the ideal generated by $(z^2, 0)$ has colength two in $\mathbb{C}\{z\}$, and perturbing it to $(z^2, t^\eta z)$ with $\eta > 0$ makes the quotient K , of length one, because t^η is a unit in K . This contradicts nothing, since [theorem 9.3](#) requires exactly n equations with an isolated complete-intersection leading zero.

Positive perturbation support. Replacing x by $x - 1$ changes the ordinary leading equation and removes its zero from the monad of the origin; an error with an exponent below zero can dominate the leading system entirely. Positive support is the mechanism making $\text{id} + T$ and $1 + \delta h$ invertible with the stated support certificates. Positivity of the perturbation support fixes the leading system; changing that system may of course change the invariant.

Well-ordering, not mere positivity. See [theorem 2.8](#): $\{1/r : r \geq 1\}$ is positive but not well ordered, and the corresponding putative Hahn series is not an element of any of our rings.

One common ordinary neighbourhood — where it is needed and where it is not. For every statement proved over $\mathcal{R}_n$ the fixed domain is not an optional analytic convenience: it is what makes [theorem 9.5](#) and [theorem 10.1](#) available at all, and individually holomorphic coefficient germs with radii tending to zero need not be coherent data in that sense. Conversely, coefficients may have arbitrarily large ordinary norms on the common domain; no unmentioned boundedness hypothesis is used. For every statement proved over $\mathcal{A}_n$ the fixed domain is *not* available, and its absence is the content of the theorem: [theorem 3.10](#) and [section 13.2](#) are elements and systems no common-domain theorem reaches.

A set-sized workspace before passing to the class. Noetherianity and finite-dimensional linear algebra are asserted for the set-sized rings $\mathcal{A}_n$ and $\mathcal{R}_n(\Gamma)$. The passage to the full surreal

class is made by the explicit base-change arguments of [section 11](#) for one finite cluster at a time, never by treating a proper class as an ordinary Noetherian ring.

A nonzero value group. [Theorem 5.6](#) shows the strong Nullstellensatz fails at $I = (0)$ when $\Gamma = \{0\}$. The hypothesis is essential rather than aesthetic, and it enters through [theorem 3.5](#) and [theorem 3.9](#).

The chosen analytic categories. These results do not classify arbitrary fine-locally analytic functions on the class plane, and they do not settle the supplied manuscripts' separate questions about essential singularities, resurgent sectorial summation, or transcendental all-scale atlases. Those concern compatibility of different expansion structures rather than a fixed positive-support deformation of an ordinary finite germ.

14.2 What is proved, and what is not

The following list is deliberately explicit. Each item is a limitation that one of the two merged manuscripts stated about its own results, and each is preserved verbatim in substance.

Status of the whole. The assertions are proved from explicitly identified classical inputs. The proposed contribution is the precise radius-free Hahn-germ package, the common-domain normal-form and residue package, and the comparison between them — not the invention of non-Archimedean Weierstrass theory or residue duality. No named published conjecture is claimed solved, no exhaustive novelty certification is asserted, and the proofs have not been formally verified. Neither a Hahn Weierstrass theorem nor a non-Archimedean Nullstellensatz is claimed as a new general idea; the coefficient ring and its support restrictions must be compared, not just theorem titles. We do not claim that our rings lie outside every broader analytic-structure framework. In the sources inspected we did not locate the exact combined statements proved here; we offer them as proposed new results in a precisely specified setting, and they may also be derivable from a broader existing framework. This is a proposed original synthesis and formulation, not a certified first theorem in multivariable non-Archimedean analysis. The article supplies proofs rather than treating a failure to locate an identical statement as proof of priority.

Scope discipline. No proper-class polynomial ring or proper-class sum is used. Every theorem about a ring concerns a fixed set-sized Γ . The support bounds are support bounds, not real norm estimates, and not a claim that successive approximations converge in the full surreal fine topology; strong summability remains the definition of the series, and the estimates [\(3.10\)](#) do not turn every sequence of partial sums into a convergent sequence.

Per-theorem non-claims.

- [Theorem 4.3](#) is a genuine finite-mapping statement with infinitesimal parameter values, not only a one-variable factorization at a single point. It does *not* assert that the roots can be globally labelled by single-valued analytic functions through a branch locus.
- [Theorem 5.3](#) gives existence of a Nullstellensatz witness, *not* a uniform bound on the complexity of finding one from arbitrary support presentations.

- The splitting (8.2) is a choice of linear splitting, *not* a claim of uniqueness of the quotients $h_j(g)$, and no ordinary norm bound on it is assumed. The same applies to the vector-space splittings (9.7): no boundedness or continuity of them is asserted.
- The classical facts of section 8.2 are used on one finite family at a time. We never assert that all coefficient germs of a Hahn perturbation admit a common contour or a common analytic representative.
- Equation (8.8) is motivated by expanding each reciprocal $(f_j + E_j)^{-1}$, but the displayed series, *not* a putative surcomplex contour integral, is the definition; and in theorem 10.1 the cycle is an ordinary one and is not treated as a fine-continuous parametrized object in $\mathbf{SC}^n$.
- The word *finite* in theorem 7.1 has its algebraic meaning throughout. No claim of properness, compactness, or classical path lifting in the full surreal fine topology is included.
- Theorem 9.4 is stated for a complex with homology only in degree zero. In a complex with homology in several degrees, positive perturbations can produce a nontrivial differential on the homology space, and the theorem would have to be modified.
- The recursion (9.14) does *not* assert a finite-time enumeration algorithm for every set-theoretically possible Hahn input; the finite-dependence statement at a specified coefficient is unconditional. More generally, the arguments are algebraic and support-theoretic rather than a complexity analysis: the set of dependencies of a fixed coefficient is finite, but an arbitrary ordered group or support need not have an effective representation from which those dependencies can be enumerated.
- Theorem 10.6 protects total length. It neither requires nor implies that the number of distinct points stays constant.
- Theorem 10.3 identifies bilinear spaces, not algebras; it does not erase the nonreduced algebraic structure of a collision, and a perfect residue pairing is not the usually degenerate trace pairing of a nonreduced finite algebra. At a multiple point (10.8) is not a valid formula with a zero denominator (theorem 10.9).
- Theorem 13.1 does not assert that the explicit matrices (13.18) remain valid for the nonpolynomial system, nor that all five of its roots are simple without computing the new Jacobian determinant.
- No Noether normalization or finite-projection theorem is claimed for $\mathcal{A}_n$ (theorem 7.2), and no analogue of theorem 11.5 is claimed for $\mathcal{R}_n$.
- The relative version of the normal-form construction over a positive-dimensional ordinary parameter space, as a module over the coherent parameter algebra rather than as complex-linear splittings on the total space, is *not* claimed: the proofs above deliberately do not assert that stronger relative linearity.
- The attainment of both valuation bounds in the Hensel–Rouché theorem, including the second-order bound, is proved in the companion report [10] and is not re-proved here (theorem 12.6).
- The symbolic checks accompanying the source manuscripts concern the displayed formulas only. They do not implement arbitrary Hahn fields and are not machine verification of the Noetherian, flatness, residue, conservation, or stability proofs.

14.3 Specific research directions left open

Four questions are left open by these proofs, and they are questions, not claims about the current status of any named conjecture.

First, whether natural common-domain versions of the radius-free construction support useful global gluing and coherent sheaf theorems without reinstating a hidden uniform-radius assumption.

The Noetherian results proved here are ring theorems on an entire monad, not gluing theorems, and neither ring is given a sheaf theory, Cartan’s theorems for surcomplex domains, or an extension of ordinary contour integration to arbitrary fine-topological paths.

Second, a weighted version of [theorem 12.2](#): the single scalar loss κ_a treats every coordinate with the same valuation threshold, and diagonal weights adapted to the Newton data may improve separation bounds for anisotropic clusters such as (13.1). [Theorem 12.5](#) prevents a universal improvement of the scalar factor two but does not preclude stronger coordinate-sensitive hypotheses.

Third, a detailed comparison with existing separated analytic structures: one should determine the exact axiomatic status of the radius-free convergent-coefficient subsystem, including uniformity properties not used here. [Theorem 11.5](#) does not by itself prove every model-theoretic or global analytic property of a larger full separated system.

Fourth, replacing an isolated complete intersection by a finite Cohen–Macaulay module with a selected free resolution. The positive conjugacy argument of [theorem 9.4](#) already applies to any complex with homology only in degree zero, provided a perturbation of its differential is supplied that still squares to zero; the genuinely new problem is to construct the required compatible perturbation of *all* syzygies rather than of the first equations only. For complete intersections the Koszul construction supplies it automatically.

A The finite-family classical input and the trace proof

We spell out the ordinary deformation statement invoked in [theorem 10.5](#). Let

$$g_i(z, u) = f_i(z) + \sum_{\ell=1}^q u_{\ell} q_{i\ell}(z), \quad q_{i\ell} \in R_n,$$

with f satisfying (8.1). The germs in this *finite* family have a common ordinary representative neighbourhood — which is exactly why the reduction of [theorem 10.4](#) is what makes the radius-free case work. The ordinary finite-mapping theorem, obtained from Weierstrass division, says that $C = \mathbb{C}\{z, u\}/(g_1, \dots, g_n)$ is a finite module over $\mathbb{C}\{u\}$ near the distinguished fibre. The sequence consisting of the g_i followed by the parameters u_{ℓ} is a system of parameters in the regular local ring $\mathbb{C}\{z, u\}$, since quotienting by it gives the finite-dimensional algebra $R_n/(f)$; regular local rings are Cohen–Macaulay, so this system of parameters is a regular sequence, and hence the parameters form a regular sequence on C . The usual finite-module flatness criterion over the regular parameter ring shows C is free over $\mathbb{C}\{u\}$ of rank μ , and its reduction modulo (u) has basis the chosen $b_1, \dots, b_{\mu}$, so those same lifts form a basis by Nakayama’s lemma. These are standard local analytic and commutative-algebra facts [12, 14, 19]. Equivalently, the relative zero space is finite over the parameter domain, is a complete intersection and hence Cohen–Macaulay, and is therefore flat over the smooth parameter base.

For small representatives the finite algebra records the entire cluster near zero, not merely a separately chosen point of a nearby fibre. Classical local duality equips it with a parameter-holomorphic residue functional, and expanding the reciprocal factors on an ordinary protecting residue cycle gives

$$\mathrm{res}_{g_u}^{\mathrm{cluster}}(H) = \sum_{k \in \mathbb{N}^n} (-1)^{|k|} \mathrm{res}_{f^{k+1}} \left(H \prod_i \left(\sum_{\ell} u_{\ell} q_{i\ell} \right)^{k_i} \right),$$

an ordinary convergent expansion for sufficiently small parameters; equivalently it is an equality of

formal parameter series, which is all the Hahn proof needs. Powers of the f_i in the denominators do not change the residue cycle class used to extract the coefficients.

The classical trace–Jacobian identity may be understood first on fibres where all points are simple, using a change of variables at each point and the decomposition of the algebra into one-dimensional factors. To include multiple fibres, augment the family by independent constant parameters in the equations; generic small target values of a finite holomorphic map in characteristic zero are regular values, and both the cluster residue and the trace matrix are holomorphic in all parameters, so equality on the regular fibres extends to all fibres and one then sets the added parameters to zero. The algebraic version is the theorem of [18]; related finite intersection deformations appear in [15].

There is a subtle point about the arbitrary splitting h in (8.2), and about the contraction (9.7). The Neumann series built from either need not converge in an ordinary analytic norm. Nevertheless, in the (u) -adic formal completion it gives a valid identity: every coefficient of total parameter degree below N uses finitely many Neumann terms, and since the completed finite algebra is free with the chosen basis those formal coordinates are unique and coincide with the Taylor coefficients of the actual analytic coordinates. This is why no continuity assertion about h is hidden in theorem 10.5. Uniqueness also follows directly from theorem 9.4, using ordinary parameter degree to form the formal inverse.

Finally, the Hahn application never needs an infinite-parameter version of this ordinary argument. At a chosen exponent, theorem 10.4 reduces everything to one of these finite families and a finite list of its coefficient identities. That reduction is the precise substitute for the common contour that the original infinite collection of germs need not possess. No interchange of a fine-topological limit with an integral is involved anywhere.

B Support accounting and proof dependencies

The dependency chain, in both settings, is

$$\begin{aligned} \text{Hahn–Neumann lemma} &\implies \text{positive operator inversion,} \\ \text{positive operator inversion} &\implies \text{division and preparation,} \\ \text{positive operator inversion} &\implies \text{Koszul conjugacy} \implies \text{fixed-basis conservation.} \end{aligned}$$

Division supplies the Noetherian and point-identification route in each ring. Classical residue theory supplies the leading perfect form and the finite-parameter trace identity, and the support calculus gives the Hahn extension. The order avoids circularity: division first proves Noetherianity, then the maximal-ideal theorem and the strong Nullstellensatz; the complete-intersection normal form proves spanning without knowing the perturbed dimension; the residue is defined independently of roots and annihilates the perturbed ideal; its ordinary leading Gram matrix proves independence and perfect duality; only then are maximal ideals interpreted as points and local dimensions summed. The trace theorem uses ordinary finite-family residue theory, not the still-to-be-proved Hahn zero count, and the local stability theorem uses the already established length-one deformation case after a rigorously allowed rescaling.

In the table, S is a source support, E a well-ordered positive perturbation support, and E_a the union of the supports of an infinitesimal displacement.

Construction	Containing exponent set, or controlling finite-dimensional fact
Coefficientwise linear operation or differentiation	A subset of the input support S
Evaluation at $c + \varepsilon$, or infinitesimal translation	$S + E_a^*$
Positive-support operator inverse on an input	$S + E^*$
Division quotient and remainder, over $\mathcal{A}_n$ or over $\mathcal{R}_n$	$S + E^*$
Normalized preparation factors	E^*
Koszul inverse $(1 + \delta h)^{-1}$, over $\mathcal{R}_n$	$S + E^*$, on one ordinary domain
Complete-intersection normal coordinates and division certificates	$S + E^*$
Perturbed residue of a numerator	$S + E^*$
Residue Gram matrix, its inverse, and the constant frame S	E^* ; the leading Gram determinant is nonzero
Multiplication constants of the fixed ordinary basis	E^* ; reduction gives the ordinary nilpotent matrices
Individual root coordinates	Algebraic over K ; <i>not</i> necessarily supported in E^*
Workspace enlargement	Same finite basis, same matrices, algebraic closedness of the original K

The support lemma proves two things for every entry: the containing set is well ordered, and only finitely many source terms contribute at one exponent. The second is what licenses regrouping, coefficientwise differentiation, and comparison of a chosen coefficient with a finite-parameter problem. A mere lower bound on valuation would provide no such licence; this is the content of [theorems 2.6](#) and [2.7](#).

C Notation

Symbol	Meaning
$\mathbf{SC} = \mathbf{No}[i]$	Surcomplex class field
$\Gamma, t^\gamma = \omega^{-\gamma}$	Nonzero divisible set-sized value group; normal-form monomial, positive exponent meaning infinitesimal
$K = K_\Gamma, K^\circ, \mathfrak{m}_K$	$\mathbb{C}((t^\Gamma))$, its valuation ring, and its ideal of infinitesimals
$v, v_H, v_{\min}, \text{red}$	Field valuation ($v(0) = +\infty$); Hahn valuation on a coefficient ring; minimum coordinate valuation; reduction
$R_n = \mathbb{C}\{z\}$	Ordinary convergent germs at the origin
$\mathcal{O}(D)((t^\Gamma))$	Fixed-polydisk ring
$\mathcal{H}_\Gamma(U), \mathcal{R}_n$	Hahn families holomorphic on one U ; their germ ring at the ordinary origin (common-domain ring)
$\mathcal{A}_n = R_n((t^\Gamma))$	Radius-free Hahn-germ ring
$\mathcal{F}_n = \mathbb{C}[[z]]((t^\Gamma))$	Formal-coefficient Hahn ring

Symbol	Meaning
E, E^*	Positive well-ordered perturbation support; the additive monoid it generates, including the empty sum
$\mathfrak{m}_a$	Evaluation ideal $(z_1 - a_1, \dots, z_n - a_n)$
μ, e_a	Global colength; local algebraic multiplicity, equal to the formal multiplicity at the actual displaced zero
B_0, B_F, A_F	Ordinary local algebra; deformed algebra over $\mathcal{A}_n$; deformed algebra over $\mathcal{R}_n$
d_f, δ, h, U	Ordinary Koszul differential, its positive perturbation, a fixed contraction, and $1 + \delta h$
NF_F, Q_j, M_H	Basis-coordinate normal form; division certificates; multiplication matrix
$\text{res}_f, \text{Res}_F, \Lambda_F, \mathcal{G}_F$	Ordinary local residue; the Hahn series residue; the contour-expansion residue over $\mathcal{R}_n$, equal to Res_F where both are defined; the residue Gram matrix
$J_F, \kappa_a, \mathcal{B}_a$	Jacobian determinant; inverse-Jacobian valuation loss; stability ball

References

- [1] Norman L. Alling, *Foundations of Analysis over Surreal Number Fields*, North-Holland Mathematics Studies **141**, North-Holland, 1987.
- [2] Alessandro Berarducci and Vincenzo Mantova, “Transseries as germs of surreal functions,” *Transactions of the American Mathematical Society* **371** (2019), 3549–3592. [arXiv:1703.01995](#).
- [3] Eduardo Cattani, Alicia Dickenstein, and Bernd Sturmfels, “Computing multidimensional residues,” in L. González-Vega and T. Recio (eds.), *Algorithms in Algebraic Geometry and Applications*, Progress in Mathematics **143**, Birkhäuser, 1996, pp. 135–164. [arXiv:alg-geom/9404011](#); [doi:10.1007/978-3-0348-9104-2_8](#). Sections 0 and 4 give the residue-duality and trace background used here.
- [4] Raf Cluckers and Leonard Lipshitz, “Fields with analytic structure,” *Journal of the European Mathematical Society* **13** (2011), 1147–1223. [doi:10.4171/JEMS/278](#); [arXiv:0908.2376](#). The arbitrary-support, common-radius coefficient families are in §3.1, Remark 3.1.4 and Example 3.3(6) of the arXiv text; related normalization and Nullstellensatz results are in §5.2, and Propositions 5.2.5–5.2.7 are close antecedents of the structure theorems proved here.
- [5] Raf Cluckers, Leonard Lipshitz, and Zachary Robinson, “Real closed fields with nonstandard and standard analytic structure,” *Journal of the London Mathematical Society* (2) **78** (2008), 198–212. [doi:10.1112/jlms/jdn024](#); [arXiv:math/0610666](#). Theorem 2.5 and Remark 2.6 give multivariable preparation and division for nonstandard analytic structures.
- [6] *Surcomplex Analysis: Infinitesimal Calculus, Hahn-Coherent Holomorphy, and Contour Theory*, supplied research manuscript, September 21, 2026. Source filename: `surcomplex_analysis(3).tex`. Its several-variable research target motivates this continuation; none of its unproved targets is used as a theorem.
- [7] *Surcomplex Analysis: Infinitesimal Calculus, Hahn-Coherent Holomorphy, and Contour Theory*, supplied unpublished manuscript, September 21, 2026. Source filename:

- `surcomplex_analysis(2).tex`. The present report uses its Hahn-coherent framework and addresses its concluding several-variable research direction. No authorship or published priority is inferred from the file.
- [8] Ovidiu Costin and Philip Ehrlich, “Integration on the surreals,” [arXiv:2208.14331](#), version 5, 2024.
 - [9] Marius Crainic, “On the perturbation lemma, and deformations,” 2004. [arXiv:math/0403266](#). Our contraction convention is $dh + hd = 1 - ip$, and [theorem 9.4](#) is proved with that sign convention.
 - [10] *Finite Hahn Deformations: Division, Conservation of Multiplicity, and Residue Duality*, companion merged report in this collection, [docs/surcomplex/finite-deformations/](#), September 21, 2026. It states the deformation package at its strongest hypotheses — arbitrary isolated complete intersection, no simple zero, no triangular system, no polynomial perturbation, no common radius — together with both stability certificates and the attainment of the valuation bounds. Its Section 15.3 is the statement of record for the lossiness of the determinant-only threshold, $0 \leq \kappa_a \leq v(\det A)$, which the present report cites rather than restates.
 - [11] *Jordan Separation, Cauchy Theory, and Stokes Calculus on the Surcomplex Plane*, companion merged report in this collection, [docs/surcomplex/contours-and-stokes/](#), September 21, 2026. It addresses the separate question of supplying a surcomplex contour that represents the series [\(8.8\)](#).
 - [12] Hans Grauert and Reinhold Remmert, *Coherent Analytic Sheaves*, Grundlehren der mathematischen Wissenschaften **265**, Springer, 1984. Classical Weierstrass division, local analytic algebras, and finite mappings.
 - [13] Phillip Griffiths and Joseph Harris, *Principles of Algebraic Geometry*, Wiley, 1978; Wiley Classics Library edition, 1994, Chapter 5, “Residues.” [doi:10.1002/9781118032527](#).
 - [14] Robert C. Gunning and Hugo Rossi, *Analytic Functions of Several Complex Variables*, Prentice-Hall, 1965; reprinted by AMS Chelsea Publishing, 2009. Ordinary Weierstrass theory, coherent analytic sheaves, and Stein-space theorems, including Cartan’s theorem B as used in [theorem 9.5](#).
 - [15] Ernst Kunz and Rolf Waldi, “Deformations of zero-dimensional intersection schemes and residues,” *Note di Matematica* **11** (1991), 247–259.
 - [16] B. H. Neumann, “On ordered division rings,” *Transactions of the American Mathematical Society* **66** (1949), 202–252. Support and well-ordering results for generalized series.
 - [17] Bjorn Poonen, “Maximally complete fields,” *L’Enseignement Mathématique* (2) **39** (1993), 87–106. [Author’s full text](#). Corollary 4 gives the algebraic closure result for divisible value group and algebraically closed residue field.
 - [18] Günter Scheja and Uwe Storch, “Über Spurfunktionen bei vollständigen Durchschnitten,” *Journal für die reine und angewandte Mathematik* **278/279** (1975), 174–190. [doi:10.1515/crll.1975.278-279.174](#).
 - [19] The Stacks Project Authors, *The Stacks Project*. [Section 10.35](#), [Jacobson rings \(tag 00FZ\)](#); [Lemma 10.97.3](#), [faithful flatness of completion \(tag 00MC\)](#); [tag 00GQ](#) for lying over; and the chapter “Commutative Algebra” for finite and integral extensions, regular local rings, Cohen–Macaulay rings and flatness. Consulted September 21, 2026.

- [20] A. K. Tsikh, *Multidimensional Residues and Their Applications*, Translations of Mathematical Monographs **103**, American Mathematical Society, 1992. [doi:10.1090/mmono/103](https://doi.org/10.1090/mmono/103).